

Construction of weights and treatment of influential units in surveys

Jean François Beaumont & David Haziza

Statistics Canada & Université de Montréal

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COURSE OUTLINE

1. Introduction
2. Sampling designs
3. Basic weighting system
4. Calibrated weighting system
5. Weighting system adjusted for nonresponse
6. Weight trimming and weight smoothing procedures
7. Why a single weighting system?
8. Influence and conditional bias
9. Robust estimation based on the conditional bias

Chapter 1: Introduction

- Surveys are used to gather data on a usually small subset of a population.
- Surveys are widely used by a variety of professions and disciplines including but not limited to national statistical offices (NSO); environmental, behavioral, and social sciences; and polling market research firms.
- Typically, the aim is to estimate finite population parameters, which are those describing some aspect of the finite population.
- Frequently encountered parameters include means, totals and proportions.
- In some cases, the interest lies in estimating more complex parameters such as quantiles and poverty indicators.

- In most surveys, information is collected on many survey variables (also called characteristics of interest) and the aim is to estimate many population parameters: such surveys are often referred to as **multipurpose surveys**.
- Most surveys conducted at NSO's are multipurpose surveys.
- Most often, the data collected are stored in a rectangular data file, each row corresponding to a sample unit (e.g., a business, a household, an individual, etc.) and each column corresponding to a survey variable (age, gender, income, etc.).
- Made available on the data file is a column of final weights. This set of weights constitutes a weighting system.

- The idea is to construct a single weighting system applicable to all the survey variables.
- With a unique weighting system, linearly weighted estimates of totals and other finite population parameters are readily obtained by data users.
- This is important for routine and timely production of statistics; see Särndal and Lundström (2005).
- Some discussions on weighting: Kish (1992), Kalton and Flores-Cervantes (2003), Särndal (2007), Gelman (2007) and its discussion, Rao et al. (2010), Haziza and Lesage (2016), Lavallée and Beaumont (2016) and Haziza and Beaumont (2017).

Sources of errors in surveys

- The estimates produced in surveys are subject to several errors.
- Let θ be a finite population parameter (e.g., a population total) and $\hat{\theta}$ be an estimate of θ .

- The total error of $\hat{\theta}$ is defined as

$$\text{total error} = \hat{\theta} - \theta.$$

- What contributes to the total error?
- Total error is the sum of sampling and non-sampling errors.

Sources of errors in surveys

Sampling errors:

- occur because the desired information is only observed for a part (the sample) of the population.
- are planned and occur deliberately.
- can be estimated from the survey data using measures such as variance, coefficient of variation or confidence interval.

Sources of errors in surveys

Non-sampling errors:

- Coverage errors: occur because of errors in the survey frame (undercoverage, overcoverage, . . .)
- Nonresponse errors: occur because the desired information is only observed for a part of the sample
- Measurement errors: occur because the observed values are different from the true values
 - ▶ can be caused by a poorly designed questionnaire, the interviewer, the respondent, etc.
- Processing errors
 - ▶ can be caused by responses not well coded, among other things

- Auxiliary information plays an important role in surveys:
 - ▶ It can be used to reduce the sampling errors → use of more efficient sampling and estimation procedures;
 - ▶ It can be used to reduce nonsampling errors such as coverage and nonresponse errors
- In the absence of nonsampling errors (ideal setup), survey statisticians try to avoid using estimation procedures whose validity depends on the validity of a given model.
- In the presence of nonsampling errors, the use of models is unavoidable. In this case, the quality of statistics depends on the quality of the assumed model, which in turns, depends on the availability of auxiliary information.

- Use of auxiliary variables at different stages:
 - ▶ The design variables: they are used at the design stage and hence, must be available for all the population units prior to sampling. They are useful for stratifying the population or in the context of proportional-to-size sampling designs.
 - ▶ The calibration (or benchmark) variables: they are used at the estimation stage. They are minimally required for all the sampled units and the corresponding population totals must be known (e.g., census counts)
 - ▶ The variables used in the treatment of nonresponse: minimally, they need to be known for all the sampled units (respondents and nonrespondents).
- A given auxiliary variable may be used at more than one stage.

Examples of auxiliary variables

- Variables on the sampling frame:
 - ▶ Household surveys: consists usually of geographical information
 - ▶ Business surveys: Geographical information (e.g., Province), NAICS (North American Industry Classification System) and size of the business
- Tax data (mainly used in business surveys)
- Demographic data (e.g., age, sex)
- Paradata: data about the survey process
 - ▶ Interviewer call record data: date and time of call, time between calls, contact strategy used.
 - ▶ Interviewer observations about each household: Type of accommodation, physical barriers, security device, indications if children present, observations about neighbourhood.

Weighting in three stages: the first stage

- Each sample unit is first assigned a design (or basic) weight, which is generally defined as the inverse of its inclusion probability.
- Most surveys use some form of unequal probability sampling design that includes stratified sampling, proportional-to-size sampling and multi-stage/phase sampling as special cases.
- Why unequal probability sampling designs?
 - ▶ To improve the statistical efficiency of the sampling strategy or to reduce the cost of sampling.
 - ▶ Subpopulations of high interest may be assigned a higher inclusion probability than other subpopulations. This is often referred to as disproportionate sampling.
 - ▶ In multi-stage designs, proportional-to-size sampling of primary sampling units may ensure self-weighting within strata. This approach provides approximately equal interviewer work loads which is desirable in terms of field operations (Rao, 2005).

Weighting in three stages: the second stage

- Virtually all surveys suffer from nonresponse, which can be defined as the failure to obtain usable responses from sample units.
- It is customary to distinguish unit nonresponse from item nonresponse.
- Unit nonresponse results in a complete lack of information on a given sample unit, whereas item nonresponse occurs when some survey variables, but not all, are collected.
- The treatment of item nonresponse is beyond the scope of this workshop.
- Response rates have been declining over time.

Weighting in three stages: the second stage

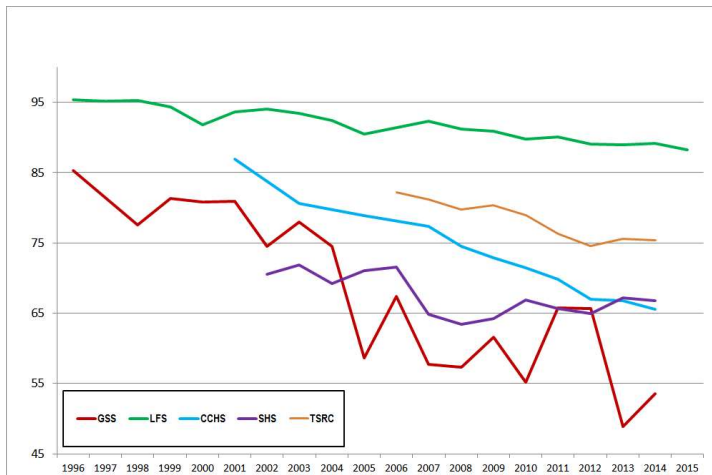


Figure 1: Nonresponse rates over time for several household surveys at Statistics Canada

Weighting in three stages: the second stage

- The most common way of dealing with unit nonresponse is to eliminate the nonrespondents from the data file and to adjust the design weights of the respondents to compensate for the elimination of the nonrespondents.
- The basic weights of respondents are multiplied by a nonresponse adjustment factor.
- At this stage, the aim is to reduce the nonresponse bias, which may be appreciable when respondents and nonrespondents are different with respect to the survey variables.
- Whether or not one will succeed in achieving an efficient bias reduction depends on the availability of powerful auxiliary information (Särndal and Lundström, 2005), which is a set of variables observed for both respondents and nonrespondents.

Weighting in three stages: the third stage

- The last stage consists of modifying the weights adjusted for nonresponse so that survey weighted estimates agree with known population totals available from external sources for important variables such as gender, age group or region.
- External sources: census, administrative data, other surveys, etc.
- This process is known as calibration.
- It includes post-stratification and raking as special cases.

Weighting: additional stage

- In some cases, the weighting process involves an additional stage during which the final weights undergo further modification.
- If the final weights are highly dispersed (because of the presence of some extreme weights) and are poorly related to the survey variables, the resulting estimators may exhibit a large variance.
- A number of approaches have been developed to improve the efficiency of survey estimates:
 - ▶ Weight trimming
 - ▶ Weight smoothing

Chapter 2: Sampling designs

Finite population parameters

- Let U be a finite population consisting of N units.
- Let $y_1, \dots, y_p$, denote p survey variables (e.g., income, age, gender, job status, etc.).
- Let $\mathbf{y}_k = (y_{1k}, \dots, y_{pk})^\top$ be the p -vector of survey variables corresponding to the k th unit, $k = 1, \dots, N$.
- We denote by $\mathbf{Y}$ the $p \times N$ matrix whose N columns are the p dimensional vectors $\mathbf{y}_1, \dots, \mathbf{y}_N$.
- A finite population parameter is any function $h(\mathbf{Y})$. Examples of finite population parameters include
 - ▶ A population total: $t_y = \sum_{k \in U} y_k$, where y is a generic notation for any survey variable.
 - ▶ A population mean: $\bar{Y} = t_y / N$.
 - ▶ A ratio of two totals: $R = \sum_{k \in U} y_{1k} / \sum_{k \in U} y_{2k}$.

Sampling design

- To estimate finite population parameters, it is common practice to select a sample S from the finite population.
- We assume that a q -vector of design variables, $\mathbf{z}$, is available for all the population units prior to sampling; i.e., available on the **sampling frame**
- Let $\mathbf{Z}$ denote a $q \times N$ matrix whose N columns are the q dimensional vectors $\mathbf{z}_1, \dots, \mathbf{z}_N$.
- In practice, virtually all surveys conducted by NSO's use some form of auxiliary information in order to stratify the population and/or to select the sample with inclusion probabilities proportional to some size variable.
- The sample S is selected according to a given sampling design $\mathcal{F}(\mathbf{I} \mid \mathbf{Z})$, where $\mathbf{I} = (I_1, \dots, I_N)^\top$ is the N -vector of sample selection indicators such that $I_k = 1$ if unit k is selected in the sample and $I_k = 0$, otherwise.

Sampling design

- Note: A sample S is completely characterized by the vector $\mathbf{I}$. In the remainder of this course, we use S and $\mathbf{I}$ interchangeably.
- Sample size: $\sum_{k \in U} I_k \rightarrow$ May be fixed (we use n) or random (we use n_S)
- The first moment of the distribution $\mathcal{F}(\mathbf{I} \mid \mathbf{Z})$:

$$E(I_k \mid \mathbf{Z}) = P(I_k = 1 \mid \mathbf{Z}) = \pi_k(\mathbf{Z}) \equiv \pi_k \quad \text{for all } k \in U.$$

- π_k : first-order inclusion probability attached to unit k .
- Property of fixed-size sampling designs: $\sum_{k \in U} \pi_k = n$.
- We assume that $\pi_k > 0$ for all $k \in U$. This condition is rarely satisfied in practice $\rightarrow$ **under-coverage**.

- The second moment of the distribution $\mathcal{F}(\mathbf{I} \mid \mathbf{Z})$ is

$$E(I_k I_\ell \mid \mathbf{Z}) = P(I_k = 1, I_\ell = 1 \mid \mathbf{Z}) = \pi_{kl}(\mathbf{Z}) \equiv \pi_{k\ell},$$

the second-order inclusion probability attached to units k and ℓ .

- Note: $\pi_{kk} = \pi_k$.
- Covariance between I_k and I_ℓ :

$$\text{Cov}(I_k, I_\ell \mid \mathbf{Z}) = \pi_{k\ell} - \pi_k \pi_\ell \equiv \Delta_{k\ell} \quad \text{for } k \neq \ell.$$

- For $k = \ell$, $\text{Cov}(I_k, I_\ell \mid \mathbf{Z}) = V(I_k) = \pi_k(1 - \pi_k)$.

Simple random sampling without replacement

- A sample S of fixed size n is a simple random sample without replacement if all the samples of size n have the same probability of being selected.
- We have $\pi_k = n/N$ for all $k \in U$.
- Here, no explicit auxiliary information is used at the design stage.

Stratified simple random sampling without replacement

- The population is partitioned into L strata $U_1, \dots, U_L$ of size $N_1, \dots, N_L$, respectively.
- We have $\bigcup_{h=1}^L U_h = U$ and $\sum_{h=1}^L N_h = N$.
- The strata are formed on the basis of auxiliary information recorded for all the population units prior to sampling (available on the sampling frame).
- From stratum h , we select a sample S_h , of size n_h , using simple random sampling without replacement. The selection in one stratum is independent from the selection in any other stratum.
- The first-order inclusion probability of unit k in stratum h is $\pi_k = n_h/N_h$.

Fixed-size inclusion probability proportional-to-size (IPPS) sampling without replacement

- We assume that a positive quantitative variable z (often called size variable) is available for all the units in the population prior to sampling (on the sampling frame).
- In IPPS designs, the units are assigned an inclusion probability, π_k , proportional to z_k . That is,

$$\pi_k = n \frac{z_k}{t_z}, \quad (1)$$

where $t_z = \sum_{k \in U} z_k$.

- Numerous fixed-size IPPS schemes have been proposed in the literature; see Tillé (2006) for a comprehensive overview of IPPS designs.

Poisson sampling

- Each unit is assigned an inclusion probability π_k .
- Poisson sampling consists of performing independent Bernoulli trials for each population unit with probability π_k , $k = 1, \dots, N$.
- If a trial is a success, the trial unit is selected in the sample S ; otherwise, it is rejected.
- Unlike the previous sampling designs, Poisson sampling is a random-size sampling design.
- IPPS Poisson sampling is obtained by setting π_k as in (1).
- Poisson sampling is useful in the context of nonresponse.

Two-phase sampling

- We discuss two-phase sampling because of its close connection with unit nonresponse.
- Two-phase sampling can be used in surveys when the sampling frame contains little or no auxiliary information.
- It consists of first selecting a large sample from the population (typically using a rudimentary sampling design) in order to collect data on variables that are inexpensive to obtain and that are related to the survey variables.
- The idea behind two-phase sampling is to create a pseudo-sampling frame richer in auxiliary information than the original frame. Then, using the variables observed in the first phase, an efficient sampling procedure can be used to select a (typically small) subsample from the first-phase sample in order to collect the survey variables.

Two-phase sampling

- Two-phase sampling may also be useful for surveying rare populations when an appropriate sampling frame does not exist.
- For instance, it is used at Statistics Canada to survey aboriginal people.
- There does not exist any frame containing aboriginal people in Canada.
- The first-phase sample consists of selecting a large sample of households and collecting basic information on all the household members such as the aboriginal status. In 2016, these data were collected through the long-form Census administered to roughly one fourth of the Canadian population.
- Then, at the second phase, a stratified sample is selected among aboriginal people selected in the first phase.

Two-phase sampling

- Two-phase sampling may be described as follows: a N -vector $\mathbf{I}_1$ is generated according to the sampling design $\mathcal{F}(\mathbf{I}_1 \mid \mathbf{Z}_1)$, where $\mathbf{I}_1$ is the N -vector of first-phase indicators and $\mathbf{Z}_1$ is a matrix of auxiliary information available for all the population units prior to first-phase sampling.
- Let S_1 be the first-phase sample. Then, a n_1 -vector of second-phase sample indicators, $\mathbf{I}_2$, is generated according to the sampling design $\mathcal{F}(\mathbf{I}_2 \mid \mathbf{I}_1, \mathbf{Z}_2^*)$, where $\mathbf{Z}_2^*$ denotes a matrix of auxiliary information available for all the first-phase sample units.
- The population version of $\mathbf{Z}_2^*$ is denoted by $\mathbf{Z}_2$ but is not observed for units not selected at the first phase.

Two-phase sampling

- Let $\pi_{1k} = P(I_{1k} = 1 \mid \mathbf{Z}_1)$ be the first-order inclusion probability of unit k in the first-phase sample.
- Let $\pi_{2k}(\mathbf{I}_1) = P(I_{2k} = 1 \mid \mathbf{I}_1, \mathbf{Z}_2^*)$ be the first-order inclusion probability of unit k in the second-phase sample.
- The second-phase sample, denoted by S_2 , is the set of population units for which both $I_{1k} = 1$ and $I_{2k} = 1$.
- Note that the inclusion probabilities at the second-phase may depend on the realized sample S_1 .
- For this reason, the inclusion probabilities π_k are intractable except when the two-phase design is invariant (Beaumont and Haziza, 2016).

Two-stage sampling

- The population U is partitioned into M clusters or primary sampling units (psu).
- At the first stage, a sample S_1 of m psu's is selected according to a given sampling design.
- At the second stage, in the i -th psu selected at the first stage, a sample S_i of secondary sampling units (ssu), also called elements or ultimate units, is selected according to a given sampling design, $i = 1, \dots, m$.
- The selection of elements in a psu selected at the first stage is independent of the selection of elements in any other psu selected at the first stage.
- The first-order inclusion probability of element k in the i th psu is given by $\pi_k = \pi_i \times \pi_{k|i}$, where π_i is the first-order inclusion probability of the i -th psu in the first-stage sample, S_1 , and $\pi_{k|i}$ is the first-order inclusion probability of element k in the i -th psu given that the i -th psu was selected in S_1 .

Two-stage sampling

- Two-stage sampling is a special case of two-phase sampling.
- **First phase**: cluster sampling (all units in the selected clusters are implicitly selected)
- **Second phase**: stratified sampling where strata are the clusters selected at the first phase
- A two-stage sampling design is an **invariant** two-phase design: the sampling design at the second stage does not depend on which clusters were selected at the first stage

Some design properties of estimators

Let θ be a finite population parameter (e.g., a population total) and let $\hat{\theta}$ be an estimator of θ .

Definition

Design expectation: $E_p(\hat{\theta}) = \sum_{s \in \Omega} \hat{\theta}(s)p(s)$,

where Ω denotes the set of all the possible samples s such that $p(s) > 0$.

The notation $E_p(\hat{\theta})$ denotes the expectation with respect to the sampling design. **Everything is treated as fixed but the vector of sample selection indicators $\mathbf{I}$.**

Definition

Design bias: $B_p(\hat{\theta}) = E_p(\hat{\theta}) - \theta$

An estimator $\hat{\theta}$ is design-unbiased for θ if $B_p(\hat{\theta}) = 0$.

Some design properties of estimators

Definition

Design variance:

$$V_p(\hat{\theta}) = E_p[\hat{\theta} - E_p\{\hat{\theta}\}]^2 = \sum_{s \in \Omega} [\hat{\theta}(s) - E_p\{\hat{\theta}\}]^2 p(s)$$

Definition

Design mean square error:

$$\begin{aligned} MSE_p(\hat{\theta}) &= E_p(\hat{\theta} - \theta)^2 \\ &= V_p(\hat{\theta}) + B_p(\hat{\theta})^2 \end{aligned}$$

$MSE_p(\hat{\theta}) = V_p(\hat{\theta})$ when $\hat{\theta}$ is design-unbiased for θ .

How to interpret design properties?

- We have a population of units (U) containing the values of the variable of interest y , for all units $k \in U$.
- Sample s is generated using a probabilistic sampling design. In practice, we draw a single sample, $s^{(1)}$, and make inferences on the basis of that sample only. The sampling error, $\hat{\theta}^{(1)} - \theta$, is unobservable because θ is unknown.
- In theory, we could draw a large number (say, R) of independent samples using the design. This would yield R sampling errors, $\hat{\theta}^{(1)} - \theta, \dots, \hat{\theta}^{(R)} - \theta$.
- The bias, $E_p(\hat{\theta} - \theta)$, is simply the average of the sampling errors as $R \rightarrow \infty$.
- The design variance and MSE can be interpreted similarly.

Chapter 3: Basic weighting system

Weighting system

- Focus: estimation of the population total, $t_y = \sum_{k \in U} y_k$
- We consider estimators of t_y of the form

$$\hat{t}_y = \sum_{k \in S} w_k y_k,$$

where w_k is a weight attached to unit k .

- The weight w_k may vary from one unit to another and, for a given unit, may vary from one sample to another.
- The set $\{w_k; k \in S\}$ is called a weighting system.
- The weights w_k are constructed without using the values of the survey variables $\rightarrow$ the weighting system $\{w_k; k \in S\}$ can be applied to any survey variable.

Basic weighting system

- Except for two-phase sampling designs, the basic (or design) weight attached to unit k is defined as the inverse of its inclusion probability. That is,

$$d_k = 1/\pi_k.$$

- d_k can be interpreted as the number of population units that sample unit k represents (**Does not always make sense!**)
- Basic weighting system: $\{d_k; k \in S\}$.
- Applying the basic weighting system to the y -variable leads to the well-known expansion estimator or π -estimator (Narain, 1951; Horvitz and Thompson, 1952) of t_y :

$$\hat{t}_{y,\pi} = \sum_{k \in S} d_k y_k = \sum_{k \in U} d_k y_k I_k. \quad (2)$$

Basic weighting system

- Noting that $E_p(I_k) = \pi_k$, the estimator (2) is design-unbiased for t_y regardless of the survey variable y being considered. We have

$$E_p(\hat{t}_{y,\pi}) = t_y \quad \text{for all } \mathbf{y} = (y_1, \dots, y_N)^\top \in \mathbb{R}^N$$

- Also, under mild regularity conditions, $\hat{t}_{y,\pi}$ is design-consistent for t_y in the sense that $\hat{t}_{y,\pi} - t_y = O_p(N/\sqrt{n})$.
- In the absence of nonsampling errors, the basic weighting system ensures that, when applied to any survey variable y , the resulting estimator is design-unbiased and design-consistent.
- **Special case:** Estimation of the population size N , which can be expressed as a total; i.e., $N = \sum_{k \in U} 1$. Therefore, a design-unbiased estimator of the population size is

$$\hat{N}_\pi = \sum_{k \in S} d_k.$$

Basic weighting system

- For a fixed-size sampling design, the expansion estimator provides a perfect estimate of the true total if $y_k = c\pi_k$ for all k for some constant c :

$$\hat{t}_{y,\pi} = \sum_{k \in S} \frac{c\pi_k}{\pi_k} = nc = c \sum_{k \in U} \pi_k = \sum_{k \in U} c\pi_k = \sum_{k \in U} y_k = t_y.$$

- Both the bias and the variance are equal to zero!
- Therefore, we expect $\hat{t}_{y,\pi}$ to be very efficient if:
 - (i) the relationship between y_k and π_k is linear and goes through the origin;
 - (ii) the coefficient of correlation between both variables is high.
- **Issue:** In practice, it is not possible to satisfy the requirement $y_k \propto \pi_k$ as the inclusion probabilities are set prior to sampling, and the y -values are available only after the sample has been selected. Also, there are usually multiple y variables.

Basic weighting system

- For simple random sampling without replacement, the design weight of a unit $k \in U$ is:

$$d_k = N/n$$

—→ self-weighting design.

- For stratified simple random sampling without replacement, the design weight of a unit $k \in U_h$ is:

$$d_k = N_h/n_h$$

- For an IPPS sampling design, the design weight of a unit $k \in U$ is:

$$d_k = \frac{t_z}{nz_k}$$

- For two-stage sampling, the design weight of a unit k in psu i is:

$$d_k = \pi_i^{-1} \times \pi_{k|i}^{-1}$$

Naive weighting system

- Why not use the SRS weighting system $\{N/n; k \in S\}$ instead (even when the design is not SRS)?
- This weighting system makes no use of the sampling design; i.e., the sampling design is discarded.
- The resulting estimator of t_y is the unweighted estimator

$$\hat{t}_{y,un} = \frac{N}{n} \sum_{k \in S} y_k = \frac{N}{n} \sum_{k \in U} y_k I_k.$$

- However, $\hat{t}_{y,un}$ is generally design-biased.

Naive weighting system

- Assuming a sampling design with a fixed sample size n , its relative bias is given by

$$\frac{E_p(\hat{t}_{y,un}) - t_y}{t_y} = \frac{(N-1)}{N} R_{y,\pi} CV(y) CV(\pi), \quad (3)$$

where

$$R_{y,\pi} = \frac{1}{(N-1)} \frac{\sum_{k \in U} (y_k - \bar{Y})(\pi_k - \bar{\pi})}{S_y S_\pi}$$

denotes the finite population coefficient of correlation between y and π with $\bar{Y} = t_y/N$, $\bar{\pi} = \sum_{k \in U} \pi_k/N$,

$$S_y = \left[(N-1)^{-1} \sum_{k \in U} (y_k - \bar{Y})^2 \right]^{1/2}$$

and S_π is similarly defined, $CV(y) = S_y/\bar{Y}$ and $CV(\pi) = S_\pi/\bar{\pi}$ are the coefficients of variation of the y -variable and the π -variable, respectively.

- We conducted a simulation study in order to compare the performance of $\hat{t}_{y,\pi}$ and $\hat{t}_{y,un}$ in terms of bias and efficiency.
- We generated a finite population of size $N = 1000$ consisting of ten variables: a size variable z available for all the population units prior to sampling and nine survey variables y_1 - y_9 .
- The nine survey variables were generated in order to illustrate the problem of the choice of weights in multipurpose surveys.
- The z -values were first generated according to a Gamma distribution with shape parameter equal to 1 and scale parameter equal to 10, resulting in $CV(z) = 1$.

Empirical study

- Given the z -values, the values of y_1 to y_9 were generated according to the model

$$y_{tk} = \beta_{0t} + \beta_{1t}z_k + \sqrt{z_k}\epsilon_k, \quad t = 1, \dots, 9$$

where the errors ϵ_k were generated from a normal distribution with mean equal to 0 and variance σ^2 .

- The values of β_{0t} and β_{1t} were chosen so that the coefficient of correlation between y_t and z , $R_{y,z}$, was set to $0.1t$, $t = 1, \dots, 9$.
- From the population, we selected 10, 000 samples of size $n = 50$ with probability proportional-to-size so that $\pi_k = nz_k / \sum_{k \in U} z_k$.
- For such a design, we have $R_{y,z} = R_{y,\pi}$ and $CV(\pi) = CV(z) = 1$. This suggests that the π_k 's were fairly dispersed in our experiment.

Empirical study

- Goal: estimate the population total for each of the characteristics of interest $y_1 - y_9$.
- In each sample, we computed $\hat{t}_{y,\pi}$ and $\hat{t}_{y,un}$.
- Define the Monte Carlo average of an estimator $\hat{t}$ by

$$E_{MC}(\hat{t}) = \frac{1}{10,000} \sum_{r=1}^{10,000} \hat{t}_{(r)},$$

where $\hat{t}_{(r)}$ denotes the estimator $\hat{t}$ in the r -th repetition, $r = 1, \dots, 10,000$.

- As a measure of bias of an estimator $\hat{t}$, we computed the Monte Carlo percent relative bias defined as

$$RB_{MC}(\hat{t}) = 100 \times \{E_{MC}(\hat{t}) - t_y\}/t_y.$$

Empirical study

- As a measure of variability of $\hat{t}$, we computed the Monte Carlo variance defined as

$$V_{MC}(\hat{t}) = E_{MC}\{\hat{t} - E_{MC}(\hat{t})\}^2.$$

- Finally, we computed the Monte Carlo mean square error defined as

$$MSE_{MC}(\hat{t}) = E_{MC}(\hat{t} - t_y)^2.$$

- Figure 2 shows the Monte Carlo percent relative bias of $\hat{t}_{y,\pi}$ and $\hat{t}_{y,un}$ for varying values of $R_{y,\pi}$.
- Figure 3 shows the percent relative variance and the percent relative efficiency

$$RV = 100 \times \frac{V_{MC}(\hat{t}_{y,\pi})}{V_{MC}(\hat{t}_{y,un})}, \quad RE = 100 \times \frac{MSE_{MC}(\hat{t}_{y,\pi})}{MSE_{MC}(\hat{t}_{y,un})}$$

for varying values of $R_{y,\pi}$.

Empirical study

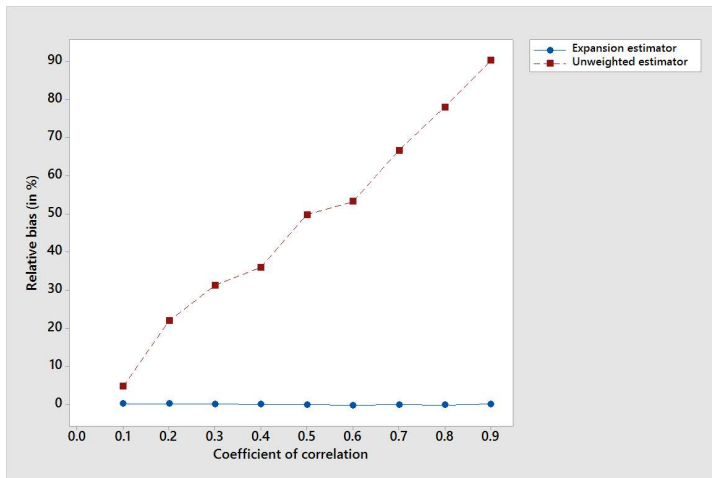


Figure 2: Relative bias of the Horvitz-Thompson and the unweighted estimator for several values of $R_{y,\pi}$

Empirical study

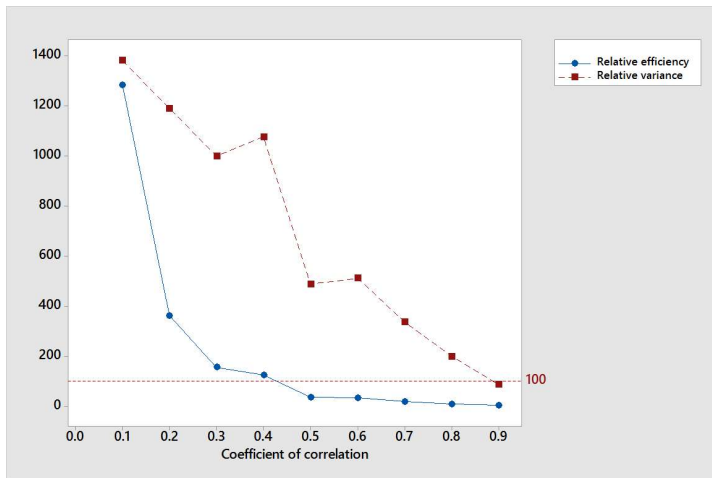


Figure 3: Relative variance and relative efficiency for several values of $R_{y,\pi}$

The Basu's elephant example

The circus owner is planning to ship his 50 adult elephants and needs a rough estimate of the total weight of the elephants. As weighing an elephant is a cumbersome process, the owner wants to estimate the total weight by weighing just one elephant. Which elephant should be weighed?

The owner looks back on his records and discovers a list of the elephants' weights taken 3 years ago. He finds that Sambo, the middle-sized elephant, was the average elephant (in weight) in his herd. He checks with the elephant trainer who reassures him (the owner) that Sambo may still be considered to be the average elephant in the herd. Therefore, the owner plans to weigh Sambo and take $50y$ (where y is the present weight of Sambo) as an estimate of the total weight $t_y = y_1 + y_2 + \dots + y_{50}$ of the 50 elephants.

The Basu's elephant example

But the circus statistician is horrified when he learns of the owner's purposive sampling plan. "How can you get an unbiased estimate of t_y this way?" protests the statistician. So, together they work out a compromise sampling plan. With the help of a table of random numbers they devise a plan that allots a selection probability of 99/100 to Sambo and equal selection probability 1/4900 to each of the other 49 elephants. Naturally, Sambo is selected and the owner is happy.

"How are you going to estimate t_y ?", asks the statistician. "Well, the estimate ought to be $50y$ of course," says the owner. "Oh! No! That cannot possibly be right," says the statistician, "I recently read an article in the Annals of Mathematical Statistics where it is proved that the expansion estimator is the unique hyper-admissible estimator in the class of the generalized polynomial unbiased estimators." "OK, What is the estimate in this case?" asks the owner, duly impressed.

The Basu's elephant example

"Since the selection probability for Sambo in our plan was $99/100$," says the statistician, "the proper estimate of Y is $100y/99$ and not $50y$."

"And, how would you have estimated t_y ," inquires the incredulous owner, "if our sampling plan made us select, say, the big elephant Jumbo? "

"According to what I understand of the expansion estimation method," says the unhappy statistician, "the proper estimate of t_y would then have been $4900y$, where y is Jumbo's weight." That is how the statistician lost his circus job (and perhaps became teacher of statistics!).

This example shows that, although the expansion estimator is design-unbiased when applied to any y -variable and for any n , it could be very inefficient in some situations. The question that comes to mind is: in Basu's example, why is the expansion estimator doing so poorly?

The Basu's elephant example

As mentioned before, the expansion estimator is efficient when the survey variable y is approximately proportional to the inclusion probabilities π . We now show that, in Basu's example, the two variables are weakly related **if the weight of Sambo is close to the average weight**.

Let y_S be the weight of Sambo. **Assume that** $y_S \approx \bar{Y}$ (i.e., the owner accurately guessed the weight of Sambo). Also, let $\pi_S = 99/100$ and $\pi_0 = 1/4900$ be the inclusion probabilities of Sambo and the other elephants, respectively. Then, the correlation $R_{y,\pi}$ between π and the variable of interest (here, weight of the elephant) is

$$R_{y,\pi} = \frac{1}{N-1} \frac{\sum_{k \in U} (\pi_k - \bar{\pi})(y_k - \bar{Y})}{S_\pi S_y} \approx \frac{1}{N-1} \frac{(\pi_S - \pi_0)(y_S - \bar{Y})}{S_\pi S_y}$$

In this example, $R_{y,\pi} \approx 0$ provided the assumption $y_S \approx \bar{Y}$ holds. Furthermore,

$$CV(\pi) = \frac{S_\pi}{\bar{\pi}} \approx 7.$$

The Basu's elephant example

The problem with the expansion estimator, in the context of the Basu's elephant example, is that

- the inclusion probabilities are most likely weakly related to the values of the variable of interest;
- the inclusion probabilities are widely dispersed (**make things worse**).

The problem could have been attenuated by using auxiliary information (elephant weights three years ago) and perhaps IPPS sampling.

Another issue is the small sample size: the advantage of probability sampling over nonprobability sampling basically disappears for a sample of size 1.

Basic weighting system: two-phase sampling

- As mentioned previously, the π_k 's are generally unknown in the case of an arbitrary two-phase design.
- Therefore, the basic weighting system $\{d_k; k \in S_2\}$ described cannot be used since the $d_k = 1/\pi_k$'s are unknown.
- Solution: we use the weighting system $\{\tilde{d}_k; k \in S_2\}$, where

$$\tilde{d}_k = \frac{1}{\pi_{1k}\pi_{2k}(I_1)}.$$

- Note that $\tilde{d}_k \neq d_k$, in general, unless the two-phase design is invariant.

Basic weighting system: two-phase sampling

- Applying the weighting system $\{\tilde{d}_k; k \in S_2\}$ to the survey variable y leads to double expansion estimator (Särndal et al., 1992, Chapter 9) of t_y :

$$\hat{t}_{y,DE} = \sum_{k \in S_2} \tilde{d}_k y_k = \sum_{k \in U} \tilde{d}_k y_k l_{1k} l_{2k}. \quad (4)$$

- The estimator (4) is design-unbiased for t_y regardless of the characteristic of interest y being considered. That is,

$$E_p(\hat{t}_{y,DE}) = E_1 E_2(\hat{t}_{y,DE} \mid I_1) = t_y \quad \text{for all } \mathbf{y} = (y_1, \dots, y_N)^\top \in \mathbb{R}^N,$$

where $E_1(\cdot)$ denotes the expectation with respect to the first-phase sampling design and $E_2(\cdot \mid I_1)$ denotes the expectation with respect to the second-phase sampling design.

Chapter 4: Calibrated weighting system

Introduction

- In practice, some auxiliary information is often available at the estimation stage. Let $\mathbf{x}_k = (x_{1k}, \dots, x_{Jk})^\top$ be a J -vector of auxiliary variables recorded for all the sample units.
- Suppose that the vector of population totals, $\mathbf{t}_x = (t_{x_1}, \dots, t_{x_J})^\top$, is available from an external source, where $t_{x_j} = \sum_{k \in U} x_{jk}$.
- While the basic weighting system ensures unbiasedness, that is, $E_p(\hat{\mathbf{t}}_{\mathbf{x}, \pi}) = \mathbf{t}_x$, it does not produce an exact estimate for each of the J auxiliary variables; i.e., $\hat{\mathbf{t}}_{\mathbf{x}, \pi} \neq \mathbf{t}_x$, in general.
- For instance, suppose that two distinct surveys collect the variable gender for each sample unit. The estimated number of males and females will almost certainly differ in both surveys. Further, these estimates will almost certainly differ from the census counts, which begs the question: which set of estimates to consider in an analysis involving the gender?
- This lack of consistency may be embarrassing in practice.

What is Calibration?

- To cope with this problem, the basic weights are modified so that survey estimates coincide with known population totals available from external sources. This process is known as calibration.
- The reasons for using calibration are three-fold:
 - (i) to force consistency of certain survey estimates to known population quantities;
 - (ii) to reduce nonsampling errors such as nonresponse errors and coverage errors;
 - (iii) to improve the precision of estimates.
- External sources include the census, administrative records and other surveys.

- We distinguish between two cases:
 - (i) $\mathbf{x}_k$ is known for every $k \in U$. This is often referred to as complete auxiliary information.
 - (ii) $\mathbf{x}_k$ is known for every $k \in S$ and the J -vector of known population totals $\mathbf{t}_x = (t_{x_1}, \dots, t_{x_J})^\top$ is known.
- Case (i) is more flexible than case (ii) because it gives considerably more freedom to structure the vector $\mathbf{x}_k$.
- Let $\tilde{w}_k$ denote the weight after calibration for unit k .
- Let d_k be the pre-calibrated weight for unit k . For now, we assume that the pre-calibrated weights correspond to the design weights although in practice, they usually correspond to the weights adjusted for nonresponse.

Formulation of the calibration problem

- We seek a calibrated weighting system $\{\tilde{w}_k; k \in S\}$ such that the weights $\tilde{w}_k$ are “as close as possible” to the design weights d_k while satisfying the J calibration constraints

$$\sum_{k \in S} \tilde{w}_k \mathbf{x}_k = \mathbf{t}_x; \quad (5)$$

see Deville and Särndal (1992) and Deville et al. (1993).

Definition

A calibration estimator of t_y is

$$\hat{t}_{y,C} = \sum_{k \in S} \tilde{w}_k y_k.$$

Formulation of the calibration problem

- There are infinitely many possible weighting systems $\{\tilde{w}_k; k \in S\}$ that satisfy the calibration constraints (5).
- The goal is to determine a set of calibrated weights $\tilde{w}_k$ that is close (in a sense that we define below) to the set of pre-calibrated weights d_k , which ensures that the resulting calibration estimator is asymptotically design-unbiased and design-consistent.
- What does close mean?
- To answer this question we need to introduce the notion of distance between two sets of weights.

Definition

Let $G(\tilde{w}_k/d_k)$ be a function such that

- $G(\tilde{w}_k/d_k) \geq 0$ and $G(1) = 0$;
- differentiable with respect to $\tilde{w}_k$;
- the derivatives $g(u) = \partial G(u)/\partial u$ are continuous;
- strictly convex.

Note: G need not be symmetric. For this reason, it is often referred to as a pseudo-distance function.

Formulation of the calibration problem

- The calibration problem may be formulated as follows: we seek a calibrated weighting system $\{\tilde{w}_k; k \in S\}$ that minimizes

$$\sum_{k \in S} \frac{d_k G(\tilde{w}_k/d_k)}{q_k}$$

subject to

$$\sum_{k \in S} \tilde{w}_k \mathbf{x}_k = \mathbf{t}_x,$$

where the coefficient q_k is a scale factor indicating the importance of unit k in the distance calculation.

- In most practical situations, the factor q_k is set to 1.
- Using the method of Lagrange multipliers, we define

$$\phi(\tilde{w}_1, \dots, \tilde{w}_n, \boldsymbol{\lambda}) = \sum_{k \in S} \frac{d_k G(\tilde{w}_k/d_k)}{q_k} - \boldsymbol{\lambda}^\top \left(\sum_{k \in S} \tilde{w}_k \mathbf{x}_k - \mathbf{t}_x \right),$$

where $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_J)^\top$ is a J -vector of Lagrange multipliers.

Formulation of calibration problem

- Differentiating $\phi(\tilde{w}_1, \dots, \tilde{w}_n, \boldsymbol{\lambda})$ with respect to $\tilde{w}_k$ and setting the derivative equal to zero, we obtain

$$\tilde{w}_k = d_k F(q_k \boldsymbol{\lambda}^\top \mathbf{x}_k), \quad (6)$$

where $F(u) = g^{-1}(u)$ is the inverse function of $g(\cdot)$.

- $F(\cdot)$ is often referred to as **the calibration function**.
- Substituting (6) in (5) leads to

$$\sum_{k \in S} d_k F(q_k \boldsymbol{\lambda}^\top \mathbf{x}_k) \mathbf{x}_k = \mathbf{t}_x. \quad (7)$$

- It remains to solve (7) for $\boldsymbol{\lambda}$, which involves a system of J equations in J unknowns.
- This can be achieved using numerical methods such as the Newton-Raphson algorithm.

Formulation of calibration problem

- Let $\hat{\lambda}$ be the solution of (7). The calibrated weight attached to unit k is given by

$$\tilde{w}_k = d_k F(q_k \hat{\lambda}^\top \mathbf{x}_k). \quad (8)$$

- The calibration weights (8) can be thought of as the product of the pre-calibrated weight, d_k , and a calibration factor, $F(q_k \hat{\lambda}^\top \mathbf{x}_k)$.
- The latter depends on:
 - ▶ the calibration function $F(\cdot)$;
 - ▶ the characteristics of unit k through q_k and $\mathbf{x}_k$;
 - ▶ the vector of estimated coefficients, $\hat{\lambda}^\top$, that can be viewed as a measure of sample imbalance.
- Applying the calibrated weighting system $\{\tilde{w}_k; k \in S\}$ to the y -variable leads to the calibration estimator

$$\hat{t}_{y,C} = \sum_{k \in S} \tilde{w}_k y_k = \sum_{k \in S} d_k F(q_k \hat{\lambda}^\top \mathbf{x}_k) y_k. \quad (9)$$

Properties of calibration estimators

- Suppose that the y -variable is linearly related to the vector of calibration variables $\mathbf{x}$ and that the relationship is perfect; that is, $y_k = \mathbf{x}_k^\top \boldsymbol{\beta}$ for all k , where $\boldsymbol{\beta}$ is a J -vector of unknown coefficients.
- In this case, $\hat{t}_{y,C}$ provides a perfect estimate of t_y since

$$\hat{t}_{y,C} = \sum_{k \in S} \tilde{w}_k y_k = \sum_{k \in S} \tilde{w}_k \mathbf{x}_k^\top \boldsymbol{\beta} = \sum_{k \in U} \mathbf{x}_k^\top \boldsymbol{\beta} = \sum_{k \in U} y_k = t_y,$$

where the third equality follows from (5).

- Therefore, we expect $\hat{t}_{y,C}$ to be efficient if there exists a strong linear relationship between the y -variable and the vector $\mathbf{x}$.

Properties of calibration estimators

- The estimator $\hat{t}_{y,C}$ is generally design-biased; that is, $E_p(\hat{t}_{y,C}) \neq t_y$.
- However, it is approximately design-unbiased. That is, $E_p(\hat{t}_{y,C}) \approx t_y$ if n is large enough. Therefore, when n is large enough, we can safely ignore the bias.
- The estimator $\hat{t}_{y,C}$ is design-consistent for t_y in the sense $\hat{t}_{y,C} - t_y = O_p(N/\sqrt{n})$.
- If n is sufficiently large, the design variance of $\hat{t}_{y,C}$ can be approximated by

$$V_p(\hat{t}_{y,C}) \approx \sum_{k \in U} \sum_{\ell \in U} \Delta_{k\ell} \frac{E_k}{\pi_k} \frac{E_\ell}{\pi_\ell},$$

where $E_k = y_k - \mathbf{x}_k^\top \mathbf{B}$ is the "census residual" attached to unit k with

$$\mathbf{B} = \left(\sum_{k \in U} \mathbf{x}_k \mathbf{q}_k \mathbf{x}_k^\top \right)^{-1} \sum_{k \in U} \mathbf{x}_k \mathbf{q}_k y_k.$$

This is a remarkable property: all the calibration estimators have the same asymptotic variance!

Some common calibration methods

- Given a vector of auxiliary variables $\mathbf{x}_k$, different calibration functions $F(\cdot)$ lead to different calibration estimators $\hat{t}_{y,C}$, in general.
- There are some exceptions to this rule; that is, there exist vectors of auxiliary variables, $\mathbf{x}_k$, that lead to the same calibration estimator regardless of $F(\cdot)$.
- These estimators include the Hájek estimator (Hájek, 1971), the ratio estimator and the post-stratified estimator. They are presented below.

Some common calibration methods

- The Hájek estimator: when the population size N is known, it is natural to require the estimated population size $\sum_{k \in S} \tilde{w}_k$ to match the true size N .
- To that end, it suffices to set $\mathbf{x}_k = q_k = 1$ for all k so that $\sum_{k \in U} \mathbf{x}_k = N$.
- The calibrated weight $\tilde{w}_k$ in (8) reduces to

$$\tilde{w}_k = d_k \frac{N}{\hat{N}_\pi},$$

where $\hat{N}_\pi = \sum_{k \in S} d_k$ denotes the estimated population size, and the calibration estimator (9) reduces to the Hájek estimator

$$\hat{t}_{y,C} = N \frac{\hat{t}_{y,\pi}}{\hat{N}_\pi}.$$

Some common calibration methods

- The ratio estimator: suppose that a quantitative variable x_k is available for all $k \in S$ and the population total of the x -values, $t_x = \sum_{k \in U} x_k$ is available.
- Setting $q_k = x_k$ in (8) leads to

$$\tilde{w}_k = d_k \frac{t_x}{\hat{t}_{x,\pi}}$$

and the calibration estimator (9) reduces to the ratio estimator

$$\hat{t}_{y,C} = \frac{\hat{t}_{y,\pi}}{\hat{t}_{x,\pi}} t_x.$$

Some common calibration methods

- The post-stratified estimator: suppose that a categorical variable (e.g., gender, age group, etc.) with J categories is available for all the sample units and that the vector of population counts

$$\mathbf{N} = (N_1, \dots, N_j, \dots, N_J)^\top$$

is available, where N_j denotes the number of units in the population belonging to the j -th category.

- Let $\mathbf{x}_k = (x_{1k}, \dots, x_{Jk})^\top$ be the J -vector attached to unit k such that $x_{jk} = 1$ if unit k belongs to the j -th category and $x_{jk} = 0$, otherwise.
- We have

$$\mathbf{t}_x = \sum_{k \in U} \mathbf{x}_k = (N_1, \dots, N_j, \dots, N_J)^\top = \mathbf{N}.$$

Some common calibration methods

- Setting q_k in (8) to q_j if unit k belongs to the j -th category leads to the post-stratified weights

$$\tilde{w}_k = d_k \frac{N_j}{\widehat{N}_{j,\pi}} \quad \text{if unit } k \text{ belongs to the } j\text{-th category,} \quad (10)$$

where $\widehat{N}_{j,\pi} = \sum_{k \in S \cap U_j} d_k$ represents the estimated number of individuals belonging to the j -th category and U_j is the set of population units belonging to the j -th category.

- Interpretation: If $N_j / \widehat{N}_{j,\pi} = 1$, then $\tilde{w}_k = d_k$, which means that no calibration adjustment was necessary since the estimated number of individuals in the j -th category matches the known count, N_j . On the other hand, if $N_j / \widehat{N}_{j,\pi} > 1$ ($N_j / \widehat{N}_{j,\pi} < 1$), then $\tilde{w}_k > d_k$ ($\tilde{w}_k < d_k$), and the weights of sample units in the j -th category are increased (decreased). This suggests that post-stratification consists of increasing (decreasing) "the visibility" of units that are under-represented (over-represented) in the sample.

Some common calibration methods

- The calibration estimator (9) reduces to the post-stratified estimator

$$\hat{t}_{y,C} = \sum_{j=1}^J \frac{N_j}{\hat{N}_{j,\pi}} \hat{t}_{jy,\pi},$$

where $\hat{t}_{jy,\pi} = \sum_{k \in S \cap U_j} d_k y_k$.

- Let n_j be the size of $S \cap U_j$, $j = 1, \dots, J$. The n_j 's are random variables, which implies that some post-strata could be very small or empty; i.e., $n_j = 0$ for some j .
- In such cases, it is recommended to use some form of collapsing in order to reduce the number of post-strata.

Some common calibration methods: the linear method

- This method is based on the generalized chi-square distance function

$$G(\tilde{w}_k/d_k) = \frac{1}{2} \left(\frac{\tilde{w}_k}{d_k} - 1 \right)^2.$$

- The resulting calibrated weights are given by

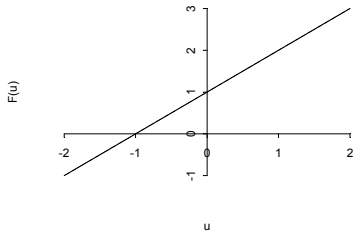
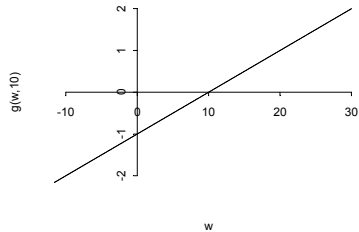
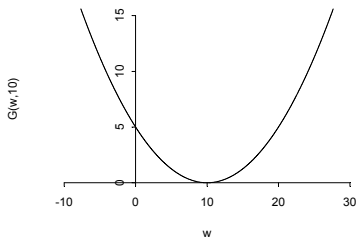
$$\tilde{w}_k = d_k(1 + q_k \hat{\boldsymbol{\lambda}}^\top \mathbf{x}_k),$$

where

$$\hat{\boldsymbol{\lambda}} = \left(\sum_{k \in S} d_k \mathbf{x}_k q_k \mathbf{x}_k^\top \right)^{-1} (\mathbf{t}_x - \hat{\mathbf{t}}_{x,\pi}).$$

- The $J \times J$ matrix $\sum_{k \in S} d_k \mathbf{x}_k q_k \mathbf{x}_k^\top$ is assumed invertible.
- With the linear method, there always exists a solution $\hat{\boldsymbol{\lambda}}$ that satisfies (7).
- This weighting method is often referred to as linear weighting.

Linear method with $d_k = 10$ and $u = q_k \hat{\lambda}^\top \mathbf{x}_k$.



Some common calibration methods: the linear method

- The calibrated weights $\tilde{w}_k$ can be written as

$$\tilde{w}_k = d_k \left\{ 1 + q_k (\mathbf{t}_x - \hat{\mathbf{t}}_{x,\pi})^\top \left(\sum_{k \in S} d_k \mathbf{x}_k q_k \mathbf{x}_k^\top \right)^{-1} \mathbf{x}_k \right\}. \quad (11)$$

- The resulting calibration estimator is given by

$$\hat{t}_{y,C} = \sum_{k \in S} \tilde{w}_k y_k = \hat{t}_{y,\pi} + (\mathbf{t}_x - \hat{\mathbf{t}}_{x,\pi})^\top \hat{\mathbf{B}}, \quad (12)$$

where

$$\hat{\mathbf{B}} = \left(\sum_{k \in S} d_k \mathbf{x}_k q_k \mathbf{x}_k^\top \right)^{-1} \sum_{k \in S} d_k \mathbf{x}_k q_k y_k.$$

- The estimator (12) is the generalized regression estimator, which has been widely studied in the context of model-assisted estimation; see Särndal et al. (1992, Chapter 6).

Some common calibration methods: the linear method

- The weights (11) may be negative. Negative weights tend to occur
 - ▶ when the size J of the vector $\mathbf{x}$ is large, which translates into a large number of calibration constraints and/or
 - ▶ when $\hat{t}_{x_j, \pi}$ is far from t_{x_j} for some $j = 1, \dots, J$.
- Although a few negative weights may have little impact on the estimates, their presence on the data file is usually deemed unacceptable by data users because a negative weight is hard to interpret.
- In some instances, negative weights could lead to negative estimates of totals or means of intrinsically non-negative variables.

Some common calibration methods: the exponential method

- This method is based on the Kullback-Leibler information distance

$$G(\tilde{w}_k/d_k) = \frac{\tilde{w}_k}{d_k} \log \frac{\tilde{w}_k}{d_k} - \frac{\tilde{w}_k}{d_k} + 1.$$

- The resulting calibrated weights are given by

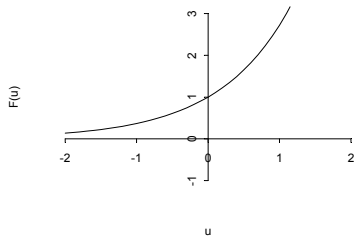
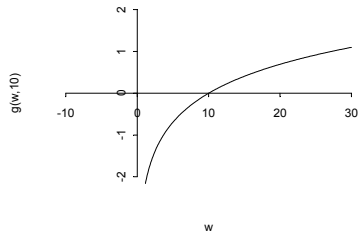
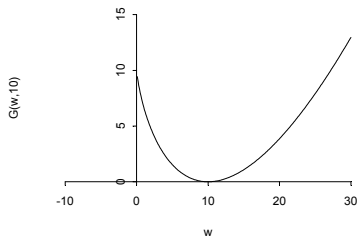
$$\tilde{w}_k = d_k \exp(q_k \hat{\lambda}^\top \mathbf{x}_k). \quad (13)$$

- Although the calibrated weights (13) are always positive, some may be extreme, which can cause the resulting estimator to be unstable.
- A solution to (7) may not exist.
- The resulting calibration estimator is

$$\hat{t}_{y,C} = \sum_{k \in S} d_k \exp(q_k \hat{\lambda}^\top \mathbf{x}_k) y_k.$$

- This weighting method is often referred to as exponential weighting.

Exponential method with $d_k = 10$ and $u = q_k \hat{\lambda}^\top \mathbf{x}_k$.



Some common calibration methods: the exponential method

- Interesting application of the exponential method: the case of two categorical calibration variables, x_1 with J_1 categories and x_2 with J_2 categories (e.g., gender and age group).
- The population U can be divided into $J_1 \times J_2$ cells, $U_{j_1 j_2}$ of size $N_{j_1 j_2}$, $j_1 = 1, \dots, J_1$; $j_2 = 1, \dots, J_2$.
- Two situations arise in practice:
 - ▶ The population counts for each cell ($j_1 j_2$), $N_{j_1 j_2}$, are known: this situation corresponds to a post-stratification based on a vector of census counts of size $J_1 \times J_2$.
 - ▶ The cell counts in the population, $N_{j_1 j_2}$, are unknown but the marginal counts, $N_{j_1 \bullet} = \sum_{j_2=1}^{J_2} N_{j_1 j_2}$ and $N_{\bullet j_2} = \sum_{j_1=1}^{J_1} N_{j_1 j_2}$ are available and can be incorporated in the estimation procedure.
- We focus on the second situation.

Some common calibration methods: the exponential method

Table 1: Population counts in the case of two categorical variables

$x_1 \setminus x_2$	1	2	...	J_2	margin
1	N_{11}	N_{12}	...	N_{1J_2}	$N_{1\bullet}$
2	N_{21}	N_{22}	...	N_{2J_2}	$N_{2\bullet}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
J_1	N_{J_11}	N_{J_12}	...	$N_{J_1J_2}$	$N_{J_1\bullet}$
margin	$N_{\bullet 1}$	$N_{\bullet 2}$	...	$N_{\bullet J_2}$	

Some common calibration methods: the exponential method

Table 2: Estimated counts in the case of two categorical variables based on the basic weighting system

$x_1 \setminus x_2$	1	2	...	J_2	margin
1	$\hat{N}_{11,\pi}$	$\hat{N}_{12,\pi}$	...	$\hat{N}_{1J_2,\pi}$	$\hat{N}_{1\bullet,\pi}$
2	$\hat{N}_{21,\pi}$	$\hat{N}_{22,\pi}$	...	$\hat{N}_{2J_2,\pi}$	$\hat{N}_{2\bullet,\pi}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
J_1	$\hat{N}_{J_11,\pi}$	$\hat{N}_{J_12,\pi}$	...	$\hat{N}_{J_1J_2,\pi}$	$\hat{N}_{J_1\bullet,\pi}$
margin	$\hat{N}_{\bullet1,\pi}$	$\hat{N}_{\bullet2,\pi}$	...	$\hat{N}_{\bullet J_2,\pi}$	

Note: $\hat{N}_{j_1j_2,\pi} = \sum_{k \in S_{j_1j_2}} d_k$, $\hat{N}_{j_1\bullet,\pi} = \sum_{j_2=1}^{J_2} \hat{N}_{j_1j_2,\pi}$ and $\hat{N}_{\bullet j_2,\pi} = \sum_{j_1=1}^{J_1} \hat{N}_{j_1j_2,\pi}$

Some common calibration methods: the exponential method

- It is virtually certain that the estimated marginal counts based on the basic design weights will not match the known population marginal counts; i.e., $\hat{N}_{j_1\bullet,\pi} \neq N_{j_1\bullet}$ and $\hat{N}_{\bullet j_2,\pi} \neq N_{\bullet j_2}$.
- How to force the consistency between the estimated margins and the population margins?
- This problem was resolved by Deming and Stephan (1940), who proposed the iterative proportional fitting procedure. The idea is to modify the estimated cell counts iteratively until the estimated margins match the population margins. The resulting estimator is the well-known raking ratio estimator.
- See the appendix for an illustration of the iterative proportional fitting procedure.

Some common calibration methods: the exponential method

- Deville et al. (1993) showed that using the exponential method in the context of two categorical variables with known margins is equivalent to the procedure of Deming and Stephan (1940).
- Let $\delta_{j_1 \bullet k}$ be the row indicator for unit k such that

$$\delta_{j_1 \bullet k} = \begin{cases} 1 & \text{if } k \text{ belongs to row } j_1 \\ 0 & \text{otherwise} \end{cases}$$

- Similarly, define the column indicator $\delta_{\bullet j_2 k}$ as

$$\delta_{\bullet j_2 k} = \begin{cases} 1 & \text{if } k \text{ belongs to column } j_2 \\ 0 & \text{otherwise} \end{cases}$$

- We define $\mathbf{x}_k = (\delta_{1 \bullet k}, \dots, \delta_{J_1 \bullet k}, \delta_{\bullet 1 k}, \dots, \delta_{\bullet J_2 k})^\top$ of size $J_1 + J_2$. The vector of known marginal population counts is given by

$$\mathbf{t}_x = \sum_{k \in U} \mathbf{x}_k = (N_{1 \bullet}, \dots, N_{J_1 \bullet}, N_{\bullet 1}, \dots, N_{\bullet J_2})^\top.$$

Some common calibration methods: the exponential method

- Let $\lambda = (a_1, \dots, a_{J_1}, b_1, \dots, b_{J_2})^\top$ be the vector of Lagrange multipliers of size $J_1 + J_2$.
- For $k \in U_{j_1 j_2}$, we have $\lambda^\top \mathbf{x}_k = a_{j_1} + b_{j_2}$.
- Setting $q_k = 1$ for all k , the calibration constraints (5) reduce to

$$\sum_{j_2=1}^{J_2} \sum_{k \in S_{j_1 j_2}} d_k \exp(a_{j_1} + b_{j_2}) = N_{j_1 \bullet}, \quad j_1 = 1, \dots, J_1$$

$$\sum_{j_1=1}^{J_1} \sum_{k \in S_{j_1 j_2}} d_k \exp(a_{j_1} + b_{j_2}) = N_{\bullet j_2}, \quad j_2 = 1, \dots, J_2$$

Some common calibration methods: the exponential method

- In this context, the calibrated weights (13) reduce to

$$\tilde{w}_k = d_k \frac{\hat{N}_{j_1 j_2, RR}}{\hat{N}_{j_1 j_2, \pi}} \text{ for } k \in S_{j_1 j_2},$$

where

$$\hat{N}_{j_1 j_2, RR} = \hat{N}_{j_1 j_2, \pi} \exp(\hat{a}_{j_1} + \hat{b}_{j_2}).$$

- The resulting calibration estimator reduces to the well-known raking ratio estimator

$$\hat{t}_{y,C} = \sum_{j_1=1}^{J_1} \sum_{j_2=1}^{J_2} \frac{\hat{N}_{j_1 j_2, RR}}{\hat{N}_{j_1 j_2, \pi}} \hat{t}_{j_1 j_2, \pi},$$

where $\hat{t}_{j_1 j_2, \pi} = \sum_{k \in S_{j_1 j_2}} d_k y_k$.

- When there are more than two categorical calibration variables, the resulting estimator is often referred to as the generalized raking estimator; e.g., Deville et al. (1993).

Some common calibration methods: the truncated linear and the logit methods

- The truncated linear and the logit methods can be used in order to ensure that the calibration adjustment factors lie between pre-specified lower and upper bounds. The use of these methods may help avoiding the occurrence of negative or extreme weights.
- The truncated linear method is based on the following distance function:

$$G(\tilde{w}_k/d_k) = \begin{cases} \frac{1}{2} \left(\frac{\tilde{w}_k}{d_k} - 1 \right)^2 & L < \frac{\tilde{w}_k}{d_k} < M \\ \infty & \text{otherwise} \end{cases}$$

for constants L and M , representing the lower and upper bounds, respectively.

- The corresponding calibration function is given by

$$F(q_k \lambda^\top \mathbf{x}_k) = \begin{cases} 1 + q_k \lambda^\top \mathbf{x}_k & (L-1) \leq q_k \lambda^\top \mathbf{x}_k \leq (M-1) \\ M & q_k \lambda^\top \mathbf{x}_k > (M-1) \\ L & q_k \lambda^\top \mathbf{x}_k < (L-1) \end{cases}$$

Some common calibration methods: the truncated linear and the logit methods

- The logit method is based on the following distance function:

$$G(\tilde{w}_k/d_k) = \begin{cases} \left(a_k \log \frac{a_k}{1-L} + b_k \log \frac{b_k}{M-1} \right) \frac{d_k}{A} & L < \frac{\tilde{w}_k}{d_k} < M \\ \infty & \text{otherwise,} \end{cases}$$

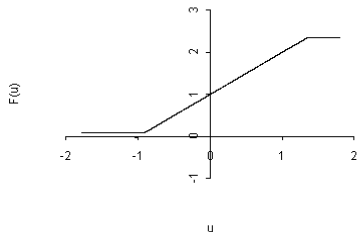
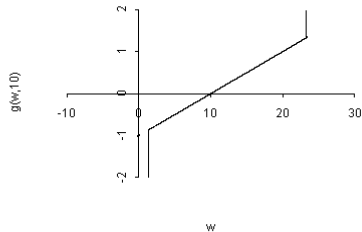
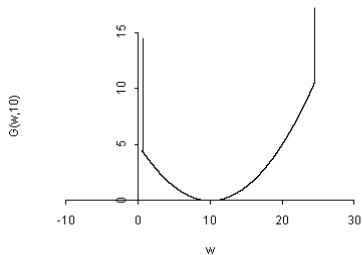
where

$$a_k = \frac{\tilde{w}_k}{d_k} - L, \quad b_k = M - \frac{\tilde{w}_k}{d_k}, \quad A = \frac{M - L}{(1 - L)(M - 1)}.$$

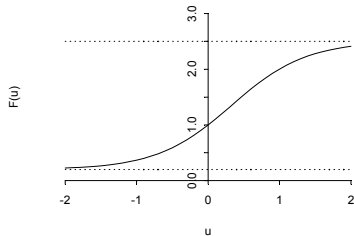
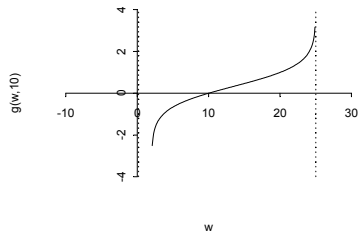
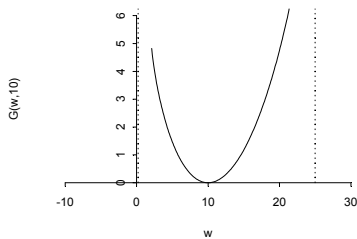
- The corresponding calibration function is given by

$$F(q_k \boldsymbol{\lambda}^\top \mathbf{x}_k) = \frac{L(M - 1) + M(1 - L) \exp(Aq_k \boldsymbol{\lambda}^\top \mathbf{x}_k)}{M - 1 + (1 - L) \exp(Aq_k \boldsymbol{\lambda}^\top \mathbf{x}_k)}.$$

Truncated linear method with $d_k = 10$, $L = 0.2$, $M = 2.5$ and
 $u = q_k \hat{\lambda}^\top \mathbf{x}_k$.



Logit method with $d_k = 10$, $L = 0.2$, $M = 2.5$ and $u = q_k \hat{\boldsymbol{\lambda}}^\top \mathbf{x}_k$.



Some common calibration methods: the truncated linear and the logit methods

- Imposing bounds on the calibration adjustment factors limits the solution space and hence lowers the possibility of the existence of a solution to the calibration equations.
- The tighter the interval the less of a chance that a solution exists.
- If an interval is too small, the weights will have a tendency to amass at the bounds.
- An examination of the distribution of $\tilde{w}_k/d_k$ is needed in order to identify the most appropriate interval for the weights.
- The larger the sample size, the more restrictive the chosen interval can be.

Some common calibration methods: summary remarks on the calibration methods

- The linear method is easy, fast and always converges but it can lead to negative weights.
- The exponential method produces positive weights but can yield extreme positive weights. It is possible that the method does not converge.
- The logit and truncated linear methods allow the weights to be bounded below and above but it is possible that these methods do not converge.

Choice of the distance function

- The calibration function is usually chosen so that the distribution of the calibration factors $\tilde{w}_k/d_k$ is “cosmetically attractive”, which we characterize as a distribution with no negative or extreme weights and that does not amass at the bounds.
- In practice, we suggest following these steps for choosing the calibration function: Start with the linear method and check whether or not there are negative calibration factors.
- If any, use the exponential method that ensures positive weights. Check whether or not the selected method has generated some extreme weights.
- If so, use any method that permits to bound the weights such as the truncated linear function or the logit function in order to ensure that the calibration adjustment factors lie between pre-specified lower and upper bounds.

Choice of calibration variables

- There are two conflicting requirements:
 - ▶ On the one hand, survey managers may be tempted to ensure consistency between survey estimates and known totals for a large vector $\mathbf{x}$ of auxiliary variables.
 - ▶ On the other hand, survey statisticians strive for efficient point estimators.
- To overcome this problem, Silva and Skinner (1997) suggested performing variable selection procedures to identify the calibration variables that are related to the characteristics of interest (or at least, a subgroup).
- An alternative is ridge calibration, whereby a cost is assigned to each calibration constraint. Assigning a large cost to a given constraint means that it is desired to stay close to the calibration constraint; see Chambers (1996), Beaumont and Bocci (2008) and Breidt and Opsomer (2016) for a discussion of ridge calibration.

Calibration weighting for two-phase sampling

- There are two levels of auxiliary information in two-phase sampling designs.
- We distinguish between two types of auxiliary variables:
 - ▶ A vector of auxiliary variables $\mathbf{x}_k^*$ of size J^* .
 - ▶ A vector of auxiliary variables $\mathbf{x}_k^o$ of size J^o .
- The availability of the auxiliary information is as follows:
 - ▶ For every $k \in S_1$, we observe $\mathbf{x}_k^o$.
 - ▶ For every $k \in S_2$, we observe both $\mathbf{x}_k^*$ and $\mathbf{x}_k^o$.
- The vector of population totals, $\sum_{k \in U} \mathbf{x}_k^*$, is known but the vector of population totals $\sum_{k \in U} \mathbf{x}_k^o$ is unknown.
- Available to us is the vector of estimated totals $\sum_{k \in S_1} \pi_{1k}^{-1} \mathbf{x}_k^o$.

Calibration weighting for two-phase sampling

- A vector of auxiliary variables, $\mathbf{x}_k$, of size $J = J^* + J^o$, is constructed by stacking the auxiliary information to obtain

$$\mathbf{x}_k = \begin{pmatrix} \mathbf{x}_k^* \\ \mathbf{x}_k^o \end{pmatrix}$$

- There exist at least three alternatives to obtain the calibrated weights $\tilde{w}_k$.
- Here, we focus on the so-called single-step calibration; see Estevao and Särndal (2006) for alternative procedures.
- Starting with the initial weights $\tilde{d}_k = 1/\{\pi_{1k}\pi_{2k}(I_1)\}$, compute the calibrated weights $\tilde{w}_k$ for $k \in S_2$ subject to the calibration constraints

$$\sum_{k \in S_2} \tilde{w}_k \mathbf{x}_k = \left(\frac{\sum_{k \in U} \mathbf{x}_k^*}{\sum_{k \in S_1} \pi_{1k}^{-1} \mathbf{x}_k^o} \right).$$

- Any calibration method (linear, exponential, etc.) can be used.

Calibration weighting for two-phase sampling

- The resulting calibration estimator of t_y is given by:

$$\hat{t}_{y,C} = \sum_{k \in S_2} \tilde{w}_k y_k.$$

- The estimator $\hat{t}_{y,C}$ is asymptotically unbiased and design-consistent for t_y .
- The material presented in this section provides a platform for single-step weighting often used for treating unit nonresponse.

Chapter 5: Weighting system adjusted for nonresponse

Introduction

- (Unit) nonresponse can be defined as the failure to obtain any usable information for some of the sample units.
- This is typically handled by deleting the nonrespondents from the survey data file and adjusting the weights of the respondents to compensate for the deletions.
- **Note:** Item nonresponse (partial information is usable) is typically handled through imputation but is outside the scope of this course.
- **Main issue with nonresponse:** bias introduced when the respondents are different from the nonrespondents with respect to the survey variables.
- Additional component of variance due to the observed sample size, n_r , that is smaller than the initial sample size, n .
- **Key to reducing both nonresponse bias and variance:** use weighting methods that take advantage of auxiliary information available for both respondents and nonrespondents.

- Unit nonresponse is often viewed as a second phase of sampling:

Two-phase sampling: $U \longrightarrow S_1 \longrightarrow S_2$

Nonresponse: $U \longrightarrow S \longrightarrow S_r$

- In the context of nonresponse, the second phase of sampling is called: the **response mechanism**
- The main difference with two-phase sampling is that the response mechanism is unknown unlike the second phase of sampling in two-phase sampling.

Response Mechanism

- Let r_k be the response indicator attached to unit k such that $r_k = 1$ if unit k is a respondent and $r_k = 0$, otherwise.
- Let $\mathbf{R}_S$ be the vector containing r_k in its k^{th} element (in two-phase sampling, we used $\mathbf{I}_2$).
- The set of respondents S_r , is the subset of S which contains all the units $k \in S$ such that $r_k = 1$.
- By analogy with sampling, we assume that the true unknown response mechanism depends only on a certain vector of variables $\mathbf{v}_k$, $k \in S$.
- Let $\mathbf{V}_S$, the matrix containing $\mathbf{v}_k^T$ in its k^{th} row.
- The response mechanism is denoted by $\mathcal{F}(\mathbf{R}_S|\mathbf{I}, \mathbf{V}_S)$.

Response mechanism

- The response probability is defined as: $p_k = P(r_k = 1 | \mathbf{l}, \mathbf{V}_S)$
- We assume that
 - ▶ $p_k = P(r_k = 1 | I_k = 1, \mathbf{v}_k)$
 - ▶ $0 < p_k \leq 1$
- The second assumption may not always be realistic because a fraction of sampled units (called the hardcore nonrespondents) may have $p_k = 0$.
- In this course, we also assume that the sample units respond independently of one another $\rightarrow$ we assume that the response indicators are mutually independent.
- This is a very common assumption but may not always hold. In that case, the dependence may be modelled but it is outside the scope of this course.

The double expansion estimator

- If p_k was known and greater than 0 for all sample units, weighting for nonresponse would be similar to weighting for two-phase sampling and the double expansion estimator of the population total t_y could be used:

$$\hat{t}_{y,DE} = \sum_{k \in S_r} \tilde{d}_k y_k = \sum_{k \in S} r_k \tilde{d}_k y_k$$

- $\tilde{d}_k = \frac{d_k}{p_k} = (\pi_k p_k)^{-1}$
- The DE estimator is q -unbiased (no nonresponse bias) in the sense that

$$B_q^{DE} = E_q(\hat{t}_{y,DE} - \hat{t}_{y,\pi} \mid \mathbf{I}) = 0$$

Properties of the double expansion estimator

- Total error = sampling error + nonresponse error

$$\hat{t}_{y,DE} - t_y = (\hat{t}_{y,\pi} - t_y) + (\hat{t}_{y,DE} - \hat{t}_{y,\pi})$$

- Bias:

$$E_{pq}(\hat{t}_{y,DE} - t_y) = E_{pq}(\hat{t}_{y,\pi} - t_y) + E_p(B_q^{DE}) = 0$$

- Total variance = sampling variance + nonresponse variance

$$V_{pq}(\hat{t}_{y,DE}) = V_p(\hat{t}_{y,\pi}) + E_p V_q(\hat{t}_{y,DE} | I)$$

- There is a price to pay for the nonresponse (in terms of an increase in variance) even in the ideal case where the bias can be eliminated.

The empirical double expansion estimator

- In practice, the p_k 's are unknown
- This problem is often addressed by determining a model for the response indicators r_k , called a response model, and then obtaining the estimated probabilities $\hat{p}_k$ using the model
- This yields the empirical double expansion estimator (or propensity score adjusted estimator) of t_y :

$$\hat{t}_{y,EDE} = \sum_{k \in S_r} \hat{d}_k y_k = \sum_{k \in S} r_k \hat{d}_k y_k$$

- $\hat{d}_k = \frac{d_k}{\hat{p}_k} = (\pi_k \hat{p}_k)^{-1}$
- Often, we have: $V_q(\hat{t}_{y,EDE} \mid I) \leq V_q(\hat{t}_{y,DE} \mid I)$
- **Why?** The estimation of response probabilities often leads to an implicit calibration of weights (as shown later), which stabilizes estimators of totals.

Modelling steps

There are two main modelling steps:

- Selection of explanatory variables that are predictive of r_k :
The goal of this step is to find $\mathbf{v}_k$ or a good proxy for $\mathbf{v}_k$
- Determination of a suitable model for the relationship between r_k and $\mathbf{v}_k$

The choice of explanatory variables highly predictive of r_k may yield:

- Small $\hat{p}_k$ and thus large weight adjustments $\hat{p}_k^{-1}$
- Unstable EDE estimator

This seems to be a paradox!

How to choose explanatory variables?

Recommendation in the literature: Explanatory variables in the vector $\mathbf{v}_k$ should be related to **both**

- the response indicator r_k and
- the variable of interest y_k (indeed, $d_k y_k$)

Explanatory variables that are related only to r_k (and not to $d_k y_k$) should be excluded for the estimation of p_k :

- Do not contribute to reducing the nonresponse bias of the EDE estimator (but do not increase it either)
- May increase substantially its nonresponse variance

Exclusion of those variables will tend to stabilize weight adjustments and the EDE estimator

Estimation of p_k under a MAR response model

- We assume that $\mathbf{v}_k, k \in S$ do not contain any missing value.
- Under this assumption, the missing y values are said to be Missing At Random (MAR).
- A parametric response model can be written as:

$$p_k = f(\mathbf{v}_k, \gamma), \quad (14)$$

for some known function $f(\mathbf{v}_k, \gamma)$, where γ is a vector of unknown model parameters.

- The estimated response probability is: $\hat{p}_k = f(\mathbf{v}_k, \hat{\gamma})$ for some estimator $\hat{\gamma}$.
- The resulting EDE estimator of t_y is approximately pq -unbiased **if the response model is correctly specified.**

Estimation of p_k under a MAR response model

- There are many possible functions f . We focus the discussion on the most common: **the logistic function**.
- With logistic regression, the response probability is modelled as:

$$p_k = f(\mathbf{v}_k, \gamma) = \frac{e^{\mathbf{v}_k^\top \gamma}}{1 + e^{\mathbf{v}_k^\top \gamma}}$$

- There are many methods for estimating γ .
- Using the Maximum Likelihood (ML) method, $\hat{\gamma}$ must satisfy the equation:

$$\sum_{k \in S} [r_k - f(\mathbf{v}_k, \hat{\gamma})] \mathbf{v}_k = \mathbf{0}$$

- Using pseudo ML:

$$\sum_{k \in S} d_k [r_k - f(\mathbf{v}_k, \hat{\gamma})] \mathbf{v}_k = \mathbf{0}$$

Estimation of p_k under a MAR response model

- Iannacchione Milne and Folsom (1991) suggested instead:

$$\sum_{k \in S} d_k \frac{[r_k - f(\mathbf{v}_k, \hat{\gamma})]}{f(\mathbf{v}_k, \hat{\gamma})} \mathbf{v}_k = \mathbf{0}$$

- **Implicit assumption:** d_k is independent of r_k after conditioning on $\mathbf{v}_k$
- Assumption satisfied when d_k is considered as a potential explanatory variable.
- **Nice calibration property** since the nonresponse-adjusted weight, $\hat{d}_k = \frac{d_k}{\hat{p}_k}$, satisfies:

$$\sum_{k \in S_r} \hat{d}_k \mathbf{v}_k = \sum_{k \in S} d_k \mathbf{v}_k$$

- Calibration constraint with benchmark totals estimated from the full sample S
- Expect low nonresponse bias and variance for y variables highly correlated with $\mathbf{v}_k$

Estimation of p_k under a MAR response model

- Unweighted version of Iannacchione Milne and Folsom (1991):

$$\sum_{k \in S} \frac{[r_k - f(\mathbf{v}_k, \hat{\gamma})]}{f(\mathbf{v}_k, \hat{\gamma})} \mathbf{v}_k = \mathbf{0}$$

- **Calibration constraint:**

$$\sum_{k \in S_r} \hat{d}_k(\pi_k \mathbf{v}_k) = \sum_{k \in S} d_k(\pi_k \mathbf{v}_k)$$

- **Should d_k be used when estimating γ ?**
- Depends on whether it is preferable to use the calibration variables $\mathbf{v}_k$ or $\pi_k \mathbf{v}_k$
- Calibration on $\mathbf{v}_k$ **may** be more meaningful in general unless the explanatory variables have already been multiplied by d_k

Estimation of p_k under a MAR response model

- Suppose no explanatory variable could be found that is related to both r_k and $d_k y_k$ so that $\mathbf{v}_k$ is chosen to be empty.
- In that case, data are said to be Missing Completely At Random (MCAR); S_r can be viewed as a random subset of S .
- The logistic model only contains an intercept ($v_k = 1, k \in S$) and it can be easily shown that the resulting $\hat{p}_k$ is:
 - ▶ **Weighted version:** $\hat{p}_k = \frac{\sum_{k \in S_r} d_k}{\sum_{k \in S} d_k}$ (weighted response rate)
 - ▶ **Unweighted version:** $\hat{p}_k = \frac{n_r}{n}$ (unweighted response rate)
- The resulting EDE estimator, sometimes called the unadjusted estimator, is approximately q -unbiased provided that the MCAR assumption holds.
- However, the nonresponse bias could be important if an explanatory variable has been omitted, which is related to both r_k and $d_k y_k$.
- Estimator is rarely (never) used in practice because the MCAR assumption is too strong.

- We generated a population of size $N = 5000$ with 7 variables: one survey variable y and 6 auxiliary variables v_1 - v_6 .
- Given v_1 - v_6 , we generated the y -variable according to the linear model

$$y_k = \beta_0 + \beta_1 v_{1k} + \beta_2 v_{2k} + \epsilon_k,$$

where the ϵ_k 's were generated from a Normal distribution with mean 0 and variance σ^2 , whose value was set so as to lead to a coefficient of determination R^2 , approximately equal to 60%.

- From the population, we selected 10,000 samples, of size $n = 200$, according to simple random sampling without replacement.

- In each sample, each unit was assigned a response propensity p_k according to the logistic function:

$$p_k = \{1 + \exp(\gamma_0 + \gamma_1 v_{1k} + \cdots + \gamma_6 v_{6k})\}^{-1}.$$

- The coefficients $\gamma_0, \gamma_1, \dots, \gamma_6$ were set so that the expected overall response rate was approximately equal to 70%.
- In each sample, the response indicators r_k were generated from a Bernoulli distribution with probability p_k .
- We were interested in estimating the population total of the y -values, $t_y = \sum_{k \in U} y_k$.
- The values of the variables v_1 - v_6 were available for all the sample units (respondents and nonrespondents).

- We considered two estimators of t_y :
 - ▶ The unadjusted estimator: $\hat{t}_{y,un} = \sum_{k \in S} d_k \frac{\sum_{k \in S_r} d_k y_k}{\sum_{k \in S_r} d_k}$
 - ▶ The EDE estimator: $\hat{t}_{y,EDE} = \sum_{k \in S_r} \frac{d_k}{\hat{p}_k} y_k$, where $\hat{p}_k$ was obtained using a logistic regression based on different subsets of v_1 - v_6 as predictors (see Table 3).
- We computed the following Monte Carlo measures:

- ▶ Monte Carlo percent relative bias:

$$RB_{MC}(\hat{t}) = \frac{1}{10,000} \sum_{k=1}^{10,000} \frac{(\hat{t}_{(k)} - t_y)}{t_y} \times 100.$$

- ▶ Monte Carlo mean square error:

$$MSE_{MC}(\hat{t}) = \frac{1}{10,000} \sum_{k=1}^{10,000} (\hat{t}_{(k)} - t_y)^2.$$

Empirical illustration: Results

Estimator	$\hat{t}_{y,un}$	$\hat{t}_{y,EDE}$ v_1	$\hat{t}_{y,EDE}$ $v_1 - v_2$	$\hat{t}_{y,EDE}$ $v_1 - v_3$	$\hat{t}_{y,EDE}$ $v_1 - v_4$	$\hat{t}_{y,EDE}$ $v_1 - v_5$	$\hat{t}_{y,EDE}$ $v_1 - v_6$	$\hat{t}_{y,EDE}$ $v_4 - v_6$
Relative bias in (%)	-3.2	-2.3	-0.1	-0.1	-0.1	-0.1	-0.1	-3.3
MSE	7.75	4.54	1.66	1.69	1.78	1.94	2.76	8.69
S_w^2	0	212	1527	2093	3465	5250	1×10^6	14993

Table 3: Monte Carlo percent relative bias and mean square error of several estimators

Note: S_w^2 denotes the Monte Carlo variability of the weights adjusted for nonresponse.

Estimation of p_k under a MAR response model

- **Main practical issues associated with the use of a parametric model**
 - ▶ The function $f(\mathbf{v}_k, \gamma)$ may not be appropriate for describing the relationship between the response indicator and the explanatory variables.
 - ▶ There may be missing interactions in the model that were not detected during model selection.
 - ▶ Predictors accounting for curvature (quadratic terms, cubic terms, etc.) may be missing.
 - ▶ Parametric models such as the logistic model may yield some estimated response probabilities, $\hat{p}_k$, that are very small resulting in very large weight adjustments $\hat{p}_k^{-1}$ and potentially unstable estimates (Recall the Basu's example).
- **Possible solution:** Form nonresponse classes homogeneous with respect to the propensity to respond using the **score method** (e.g., Little (1986); Eltinge and Yansaneh (1997) and Haziza and Beaumont (2007))

Estimation of p_k under a MAR response model

- The score method can be implemented using the following steps:
 - ▶ **Step 1:** Obtain preliminary estimated response probabilities, $\hat{p}_k^{LR}$, $k \in S$, from a logistic regression.
 - ▶ **Step 2:** Form a certain number of classes homogeneous with respect to $\hat{p}_k^{LR}$. Two methods can be considered:
 - Equal quantile method: it consists of ordering the sample from the lowest $\hat{p}_k^{LR}$ to the largest and then form equal size classes.
 - Clustering method: Use a clustering algorithm such as the k -means algorithm.
 - ▶ **Step 3:** Compute the final estimated response probability $\hat{p}_k$, for all units k in class c , as the weighted (or unweighted) response rate within class c .

Estimation of p_k under a MAR response model

- **Choice of the number of classes:** as small as possible but large enough to capture most of the variability of $\hat{p}_k^{LR}$, $k \in S$.
- Provides some robustness to model misspecifications because the observed response rate is used to estimate p_k within a given class and not the average of $\hat{p}_k^{LR}$ (**relies less heavily on the logistic regression model**).
- Less prone to extreme weight adjustments, especially if a minimum number of respondents within each class is specified.
- The nonresponse bias of the resulting EDE estimator is 0 if all sample units in a given class have all the same response probability.
- This justifies why classes are made homogeneous with respect to $\hat{p}_k^{LR}$ with the score method.

Estimation of p_k under a MAR response model

- Classes homogeneous with respect to the y variable also tend to reduce the nonresponse bias but there may be many y variables...
- Ideally, classes are homogeneous with respect to both the response probability and y variables in order to reduce both nonresponse bias and variance.
- The score method is one method of forming classes. Other methods are:
 - ▶ Cross classification of many categorical variables (somewhat ad hoc but may be reasonable if only a few explanatory variables are available)
 - ▶ The CHi square Automatic Interaction Detection (CHAID) algorithm developed by Kass (1980).
 - ▶ Regression trees; e.g., Phipps and Toth (2012).

Estimation of p_k under a MAR response model

- Nonparametric methods are an alternative to forming classes.
- **Nonparametric estimation:** $p_k = f(\mathbf{v}_k)$ and the function f does not need to be explicitly specified.
- Two nonparametric methods available in the literature are:
 - ▶ Kernel regression: e.g., Giommi (1984) and Da Silva and Opsomer (2006) $\Rightarrow$ can be viewed as moving nonresponse classes
 - ▶ Local polynomial regression: DaSilva and Opsomer (2008).
- Nonparametric methods protect (to some extent) against model misspecifications such as the non-inclusion of predictors accounting for curvature or interactions.
- These nonparametric methods typically work well when the number of explanatory variables is small.

Estimation of p_k under a MNAR response model

- **Recall:** Missing y values are MAR when v_k is observed for all $k \in S$.
- **Missing y values are said to be Missing Not At Random (MNAR)** when some (or all) of the explanatory variables in v_k are subject to missing values.
- This would occur when variables that explain response are indeed survey variables of interest (y variables).
- For example, in a survey on income, people refusing to answer the survey might be those who have a low or high income.
- Methods for handling MNAR typically require modelling both the response indicator and the y variable, and are typically complex.
- We focus on generalized calibration (e.g., Deville, 1998; Sautory, 2003; Kott, 2006), which does not require modelling the y variable and is less complex but it requires choosing **instrumental variables**.

Estimation of p_k under a MNAR response model

- Suppose that $\mathbf{v}_k^\top = (\mathbf{v}_{1k}^\top, \mathbf{v}_{2k}^\top)$
 - ▶ $\mathbf{v}_{1k}^\top$ is observed for all $k \in S$
 - ▶ $\mathbf{v}_{2k}^\top$ is observed only for $k \in S_r$ (some of the survey variables)
- The logistic function can again be used to model p_k :
 $p_k = f(\mathbf{v}_k, \gamma)$
- Estimation of γ is more difficult because of missing values in $\mathbf{v}_k$
- Generalized calibration solves this problem by substituting $\mathbf{v}_{2k}$ with a vector of instruments, $\mathbf{v}_{2k}^I$, **observed for all sample units $k \in S$** .
- Full vector is denoted by: $\mathbf{v}_k^{\prime\top} = (\mathbf{v}_{1k}^\top, \mathbf{v}_{2k}^{\prime\top})$

Estimation of p_k under a MNAR response model

- The instrumental variables $\mathbf{v}_{2k}^I$ should be:
 - ▶ observed for all $k \in S$;
 - ▶ associated as much as possible with their corresponding variables in $\mathbf{v}_{2k}$ (**for efficiency considerations**);
 - ▶ independent of r_k after conditioning on $\mathbf{v}_k$ (**for validity considerations**).
- The latter is the key assumption to ensure $\hat{p}_k = f(\mathbf{v}_k, \hat{\gamma})$ is approximately q -unbiased for p_k .
- **How to estimate γ ?** Generalized calibration is based on an estimating equation very similar to the one proposed by Iannacchione et al. (1991):

$$\sum_{k \in S} d_k \frac{[r_k - f(\mathbf{v}_k, \hat{\gamma})]}{f(\mathbf{v}_k, \hat{\gamma})} \mathbf{v}_{2k}^I = 0$$

- Note that the missing values $\mathbf{v}_{2k}$, for the nonrespondents $k \in S - S_r$, are not used.

Estimation of p_k under a MNAR response model

- The nonresponse-adjusted weight, $\hat{d}_k = \frac{d_k}{\hat{p}_k}$, satisfies the calibration equation:

$$\sum_{k \in S_r} \hat{d}_k \mathbf{v}'_k = \sum_{k \in S} d_k \mathbf{v}'_k$$

- $\mathbf{v}_k$: Explanatory variables in the response model; some are not observed for the nonrespondents.
- $\mathbf{v}'_{2k}$: **Instrumental variables but are also calibration variables (like $\mathbf{v}'_{1k}$).**
- Expect low nonresponse bias and variance for y variables highly correlated with $\mathbf{v}'_k$
- **Remark:** Some authors (e.g., Deville, 1998; Estevao and Särndal, 2000; Kott, 2006) defined $\mathbf{v}_k$ as the vector of instrumental variables. Our terminology for $\mathbf{v}_k$ better reflects its role and is in line with the econometric literature.

Estimation of p_k under a MNAR response model

- Generalized calibration (and other methods for MNAR data) is rarely used in practice. **Why?**
 - ▶ More complex and lack of software.
 - ▶ Extreme weight adjustments may occur with the logistic function or it may be misspecified.
 - ▶ Seems impossible to validate the logistic function through model diagnostics because $\mathbf{v}_{2k}$ is not observed for the nonrespondents.
 - ▶ For the same reason, a score method cannot be used to obtain robustness or reduce extreme weight adjustments.
 - ▶ Appropriate instrumental variables may not be easy to find in practice (**should be independent of r_k after conditioning on $\mathbf{v}_k$**).
- However, there are cases where it may be reasonable to assume that r_k depends directly on y_k (e.g. when y is a sensitive variable). In those cases, failure to account for that dependence might lead to significant nonresponse bias.

Calibration after nonresponse adjustment

- So far, our objective was to reduce nonresponse bias (and also variance) by using explanatory variables, $\mathbf{v}_k$, associated with both r_k and $d_k y_k$.
- There might be other auxiliary variables, unrelated to r_k , that could be useful to reduce further the nonresponse variance (and the sampling variance) through calibration weighting.
- Similarly to two-phase sampling, there are two types of auxiliary variables for calibration:
 - ▶ $\mathbf{x}_k^*$: observed at least for all $k \in S_r$
 - ▶ $\mathbf{x}_k^o$: observed for all $k \in S$ (but not for all $k \in U - S$)
- The vector of population totals $\sum_{k \in U} \mathbf{x}_k^*$ is assumed to be known.
- The vector of population totals $\sum_{k \in U} \mathbf{x}_k^o$ is unknown but replaced with an estimate from the full sample: $\sum_{k \in S} d_k \mathbf{x}_k^o$.

Calibration after nonresponse adjustment

- The vector of auxiliary variables, $\mathbf{x}_k$, is defined as:

$$\mathbf{x}_k = \begin{pmatrix} \mathbf{x}_k^* \\ \mathbf{x}_k^o \end{pmatrix}$$

- The nonresponse-adjusted weights $\hat{d}_k = d_k/\hat{p}_k, k \in S_r$, are not calibrated:

$$\sum_{k \in S_r} \hat{d}_k \mathbf{x}_k \neq \begin{pmatrix} \sum_{k \in U} \mathbf{x}_k^* \\ \sum_{k \in S} d_k \mathbf{x}_k^o \end{pmatrix}$$

- To cope with this problem, we seek a calibrated weighting system $\{\tilde{w}_k; k \in S_r\}$ that minimizes

$$\sum_{k \in S_r} \frac{\hat{d}_k G(\tilde{w}_k/\hat{d}_k)}{q_k}$$

subject to the calibration equation:

$$\sum_{k \in S_r} \tilde{w}_k \mathbf{x}_k = \begin{pmatrix} \sum_{k \in U} \mathbf{x}_k^* \\ \sum_{k \in S} d_k \mathbf{x}_k^o \end{pmatrix}$$

Calibration after nonresponse adjustment

- The resulting calibrated weights are given by

$$\tilde{w}_k = \hat{d}_k F(q_k \hat{\boldsymbol{\lambda}}^\top \mathbf{x}_k) = d_k \times \frac{1}{\hat{\rho}_k} \times F(q_k \hat{\boldsymbol{\lambda}}^\top \mathbf{x}_k). \quad (15)$$

- Any calibration method (linear, exponential, etc.) can be used.
- Applying the calibrated weighting system $\{\tilde{w}_k; k \in S\}$ to the y -variable leads to the calibration estimator

$$\hat{t}_{y,C} = \sum_{k \in S_r} \tilde{w}_k y_k \quad (16)$$

- We expect calibration to reduce both sampling and nonresponse variance of the EDE estimator if there is a strong linear relationship between the y -variable and the vector $\mathbf{x}$.

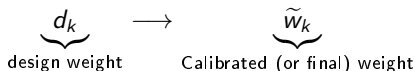
Nonresponse weighting: two approaches

- Two weighting approaches can be used for treating unit nonresponse:

- Weighting using the inverse of the estimated response probability followed by calibration:



- Calibration weighting:



- The first approach (discussed above) is often called the two-step approach, whereas the second (discussed below) is often called the one-step approach.
- In both cases, the weights of respondents are increased to compensate for the elimination of the nonrespondents.

Calibration weighting: one-step approach

- In the one-step approach, we seek a calibrated weighting system $\{\tilde{w}_k; k \in S_r\}$ that minimizes

$$\sum_{k \in S_r} \frac{d_k G(\tilde{w}_k/d_k)}{q_k}$$

subject to the calibration equation:

$$\sum_{k \in S_r} \tilde{w}_k \mathbf{x}_k = \left(\frac{\sum_{k \in U} \mathbf{x}_k^*}{\sum_{k \in S} d_k \mathbf{x}_k^o} \right)$$

- **Note:** There are again two types of auxiliary variables: $\mathbf{x}_k^*$ and $\mathbf{x}_k^o$
- Similar to the two-step approach with the exception that the initial weight is d_k instead of the nonresponse-adjusted weight $\hat{d}_k$.

Calibration weighting: one-step approach

- The calibrated weights are given by:

$$\tilde{w}_k = d_k \times F(q_k \hat{\lambda}_r^\top \mathbf{x}_k)$$

- The resulting estimator is:

$$\hat{t}_{y,C} = \sum_{k \in S_r} \tilde{w}_k y_k \quad (17)$$

- Any calibration method (linear, exponential, etc.) can be used.
- Lundström and Särndal (1999) and Särndal and Lundström (2005) argue:
"No explicit response model is needed, in contrast of the conventional methods"
- **Can we always avoid modelling p_k ?**

Calibration weighting: one-step approach

- **In the two-step approach:**

- ▶ The role of $\mathbf{v}_k$ is to reduce nonresponse bias (and nonresponse variance). This is achieved by choosing $\mathbf{v}_k$ related to both r_k and $d_k y_k$.
- ▶ The function $f(\cdot)$ is chosen so as to describe the relationship between r_k and $\mathbf{v}_k$. Model validation techniques can be used to help determine a suitable function.
- ▶ The role of $\mathbf{x}_k$ is to reduce the sampling and nonresponse variance. This is achieved by choosing $\mathbf{x}_k$ (linearly) related to y_k .

- **In the one-step approach:**

- ▶ The role of $\mathbf{x}_k$ is to reduce the nonresponse bias as well as the sampling and nonresponse variance. This can be achieved by choosing $\mathbf{x}_k$ (linearly) related to y_k .

Some more remarks on the nonresponse bias of calibration estimators

- If a linear model holds between y and x and sampling and nonresponse are **ignorable** with respect to the model, then it can be shown that the nonresponse bias of any calibration estimator vanishes in the sense that

$$E_{mpq} \left(\sum_{k \in S_r} \tilde{w}_k y_k - \sum_{k \in U} y_k \right) = 0$$

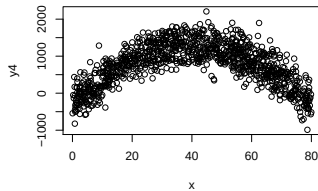
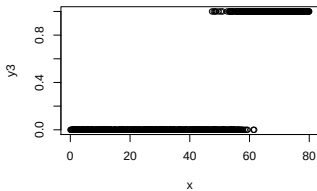
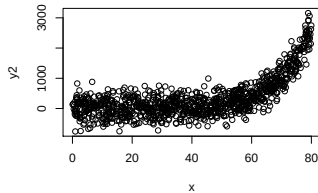
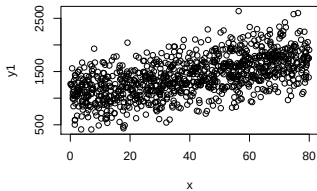
- In practice, there are many y variables and a linear model may not hold for all of them.
- **How can the nonresponse bias be controlled when there are multiple y variables?**

Calibration weighting: one-step approach

- The two-step approach takes care of the bias in the first step through explicit modelling of the response mechanism: **only r_k requires to be modelled even if there are multiple y variables.**
- The one-step approach could also take care of the bias for all y variables if the function $F(\cdot)$ is chosen so that $F(q_k \hat{\lambda}_r^\top \mathbf{x}_k)^{-1}$ is a q consistent estimate of $p_k \Rightarrow$ **modelling the response mechanism is not avoided.**
- See Haziza and Lesage (2016) for greater discussion on the nonresponse bias.
- We prefer the two-step approach because it clearly separates the weight adjustment aiming at reducing the nonresponse bias from the weight adjustment aiming at reducing variance: **Different choices of calibration variables and function can be made depending on the objective.**

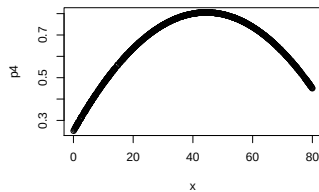
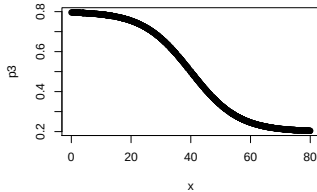
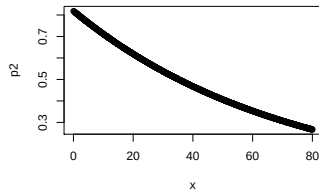
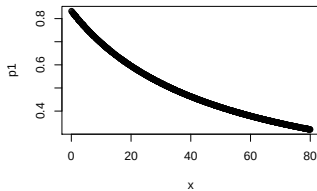
- **Goal:** Evaluate the effect of choosing an incorrect calibration function F (or function f)
- We generated a population of size $N = 1,000$, with an auxiliary variable x and 4 survey variables y_1 , y_2 , y_3 and y_4 .
- The x -values were generated from a uniform distribution $(0, 80)$.
- Relationship between y and x :
 - ▶ y_1 and x : linear
 - ▶ y_2 and x : exponential
 - ▶ y_3 and x : logistic
 - ▶ y_4 and x : quadratic
- We considered the census case; i.e., $n = N = 1000$.

Relationship between y and x



- In each population, we generated nonresponse to item y according to 4 response mechanisms with an overall expected response rate of 50%.
 - ▶ p_1 : inverse linear $\rightarrow p_k = (1.2 + 0.024 x_k)^{-1}$.
 - ▶ p_2 : exponential $\rightarrow p_k = \exp(-0.2 - 0.014x_k)$.
 - ▶ p_3 : logistic $\rightarrow p_k = 0.2 + 0.6 \{1 + \exp(-5 + x_k/8)\}^{-1}$.
 - ▶ p_4 : quadratic $\rightarrow p_k = 0.7 + 0.45 (x_k/40 - 1)^2 + 0.0025 x_k$.
- We used $R = 5000$ iterations.

Nonresponse mechanisms



Simulation study

- Goal: estimate t_{y_1} , t_{y_2} , t_{y_3} and t_{y_4} .
- We computed three estimators of t_{y_j} , $j = 1, \dots, 4$.
 - (i) The unadjusted estimator:

$$\hat{t}_{y_j, un} = \sum_{k \in S} d_k \frac{\sum_{k \in S_r} d_k y_k}{\sum_{k \in S_r} d_k}$$

- (ii) Several one-step calibration estimators:

$$\hat{t}_{y_j, C} = \sum_{k \in S_r} d_k \times F(q_k \hat{\boldsymbol{\lambda}}_r^\top \mathbf{x}_k) y_{jk} = \sum_{k \in S_r} d_k \frac{1}{\widehat{\rho}_k} y_{jk}$$

based on several calibration functions $F(\cdot)$, with $\mathbf{x}_k = (1, x_k)^\top$.

- (iii) The PSA estimator:

$$\hat{t}_{y_j, PSA} = \sum_{k \in S_r} d_k \frac{1}{\widehat{\rho}_k} y_{jk},$$

where $\widehat{\rho}_k$ was obtained using the score method with 20 classes.

Monte Carlo measures:

- Monte Carlo relative bias (in %):

$$RB_{MC}(\hat{t}) = 100 \times \frac{1}{5,000} \sum_{r=1}^{5,000} \frac{(\hat{t}_{(r)} - t_y)}{t_y}.$$

- Monte Carlo relative root mean square error:

$$RRMSE_{MC}(\hat{t}) = 100 \times \frac{\left\{ \frac{1}{5,000} \sum_{r=1}^{5,000} (\hat{t}_{(r)} - t_y)^2 \right\}^{1/2}}{t_y}.$$

Simulation study: Results

	$\hat{t}_{y,un}$	$\hat{t}_{y,C}$ $F(u) = 1 + u$	$\hat{t}_{y,C}$ $F(u) = \exp(u)$	$\hat{t}_{y,C}$ $F(u) = \text{logit}$	$\hat{t}_{y,PSA}$
y_1 (lin.)	-4.1 (4.2)	0.0 (0.7)	0.0 (0.7)	0.0 (0.7)	-0.0 (0.8)
y_2 (exp.)	-28.1 (28.7)	-0.1 (5.5)	2.8 (6.1)	3.3 (6.4)	-0.1 (3.0)
y_3 (logit)	-27.5 (27.9)	-0.1 (3.4)	1.7 (3.6)	2.1 (3.8)	-0.1 (2.3)
y_4 (quad.)	-4.8 (5.3)	0.1 (2.8)	-2.0 (3.3)	-2.4 (3.5)	-0.1 (1.4)

Table 4: p_k -inverse linear: $p_k = (1.2 + 0.024 x_k)^{-1}$

Simulation study: Results

	$\hat{t}_{y,un}$	$\hat{t}_{y,C}$ $F(u) = 1 + u$	$\hat{t}_{y,C}$ $F(u) = \exp(u)$	$\hat{t}_{y,C}$ $F(u)=\text{logit}$	$\hat{t}_{y,PSA}$
y_1 (lin.)	-4.9 (4.9)	-0.0 (0.8)	0.0 (0.8)	0.0 (0.8)	-0.0 (0.8)
y_2 (exp.)	-35.1 (35.5)	-4.0 (7.1)	-0.0 (5.8)	0.7 (5.9)	-0.1 (3.2)
y_3 (logit)	-33.8 (34.1)	-2.5 (4.3)	0.0 (3.3)	0.6 (3.3)	-0.1 (2.3)
y_4 (quad.)	-3.6 (4.3)	2.9 (4.2)	0.0 (2.7)	-0.6 (2.8)	-0.0 (1.6)

Table 5: p_k -exponential: $p_k = \exp(-0.2 - 0.014x_k)$

Simulation study: Results

	$\hat{t}_{y,un}$	$\hat{t}_{y,C}$ $F(u) = 1 + u$	$\hat{t}_{y,C}$ $F(u) = \exp(u)$	$\hat{t}_{y,C}$ $F(u)=\text{logit}$	$\hat{t}_{y,PSA}$
y_1 (lin.)	-7.3 (7.3)	-0.3 (0.9)	-0.2 (0.9)	-0.2 (0.9)	-0.1 (0.9)
y_2 (exp.)	-51.5 (51.7)	-10.0 (12.3)	-0.4 (7.0)	0.9 (7.1)	-0.2 (3.7)
y_3 (logit)	-53.4 (53.5)	-12.1 (12.9)	-5.6 (6.7)	-4.5 (5.8)	-0.3 (3.0)
y_4 (quad.)	-1.0 (2.3)	11.7 (12.3)	4.3 (5.3)	3.1 (4.3)	-0.0 (1.8)

Table 6: p_k -logistic: $p_k = 0.2 + 0.6 \{1 + \exp(-5 + x_k/8)\}^{-1}$

Simulation study: Results

	$\hat{t}_{y,un}$	$\hat{t}_{y,C}$ $F(u) = 1 + u$	$\hat{t}_{y,C}$ $F(u) = \exp(u)$	$\hat{t}_{y,C}$ $F(u)=\text{logit}$	$\hat{t}_{y,PSA}$
y_1 (lin.)	1.3 (1.4)	-0.2 (0.5)	-0.2 (0.5)	-0.2 (0.5)	0.0 (0.5)
y_2 (exp.)	-8.3 (9.4)	-19.7 (19.9)	-19.2 (19.5)	-19.0 (19.3)	-0.4 (2.3)
y_3 (logit)	-0.5 (3.2)	-11.8 (12.0)	-11.5 (11.7)	-11.4 (11.6)	-0.1 (1.0)
y_4 (quad.)	13.1 (13.2)	13.7 (13.8)	13.4 (13.5)	13.2 (13.4)	0.2 (1.1)

Table 7: p_k -quadratic: $p_k = 0.7 + 0.45 (x_k/40 - 1)^2 + 0.0025 x_k$

Chapter 6: Weight trimming and weight smoothing procedures

Weight trimming

- Units exhibiting a large weight are deemed harmful because they lead to highly dispersed weights, which can potentially cause the estimators to be unstable.
- Rationale behind weight trimming: modify the survey weights so that the resulting estimators have a lower mean square error than that of commonly used estimators (e.g., expansion estimators, calibration estimators, etc.).
- This is generally achieved at the expense of introducing a bias.
- A number of weight trimming methods have been considered in the literature; see Potter (1988, 1990).

Weight trimming

- All methods rely on a specification of a threshold, w_0 above which a unit is identified as problematic.
- Idea: to force the weights above this threshold to this value.
- Weight trimming can be used at any stage of the weighting process, although it is common to apply it on the final weighting system.
- Let w_k and $w_{k,t}$ be the weights associated with unit k before and after trimming, respectively.
- We have

$$w_{k,t} = \begin{cases} w_0 & \text{if } w_k \geq w_0 \\ \gamma w_k & \text{if } w_k < w_0 \end{cases} \quad (18)$$

where γ is a rescaling factor that ensures that

$$\sum_{k \in S} w_{k,t} = \sum_{k \in S} w_k.$$

Weight trimming

- Rationale behind the rescaling: the excess weight is redistributed among the units below the threshold so that the sum of the weights after trimming is still a valuable estimate of the population size.
- The resulting trimmed estimator of t_y is

$$\hat{t}_{y,trim}(w_0) = \sum_{k \in S} w_{k,t} y_k.$$

- If $w_0 = \infty$, then $w_{k,t} = w_k$ for all k and $\hat{t}_{y,trim}(w_0)$ reduces to the untrimmed estimator.
- The choice of the threshold w_0 is very important because a bad choice may potentially lead to a trimmed estimator with a mean square error larger than that of the untrimmed estimator.
- Ideally, the threshold should be function of the sample size n so that

$$\lim_{n \rightarrow \infty} w_{0n} = \infty.$$

Weight trimming

- In practice, the threshold w_0 is often chosen independently of the sample size; for example, it is customary to examine the distribution of the weights and to use an outlier detection method to determine the threshold. In this case, the resulting estimators are inconsistent.
- To obtain a trimmed estimator that is design consistent, we could determine the value of w_0 that minimizes the estimated mean square error of the trimmed estimator:

$$\widehat{MSE}(\hat{t}_{y,trim}) = \hat{V}(\hat{t}_{y,trim}) + (\hat{t}_{y,trim} - \hat{t}_{y,\pi})^2 - \hat{V}(\hat{t}_{y,trim} - \hat{t}_{y,\pi}).$$

- **Optimal trimmed estimator:** $\hat{t}_{y,trim}(w_0^{opt})$.
- Drawbacks:
 - ▶ Minimizing the estimated mean square error is not an easy task, in general.
 - ▶ An optimal threshold w_0^{opt} for a given survey variable y may be far from optimal for another survey variable.

Weight trimming

- A popular weight trimming method in practice consists of
 - (1) setting a value $w_0 = c\bar{w}$, where $\bar{w} = n^{-1} \sum_{k \in S} w_k$, where c is a constant whose value is set by the survey statistician.
 - (2) define the trimmed weights $w_{k,t}$ as in (18);
 - Step 2 is repeated until all the weights $w_{k,t}$ are smaller than w_0 .
- If $c = 1$, the estimator $\hat{t}_{y,trim}(w_0)$ is essentially equivalent to the unweighted estimator, $\hat{t}_{y,trim}(w_0) = \frac{N}{n} \sum_{k \in S} y_k$.

Example

Consider the set of sampling weights:

$w_1 = 1, w_2 = 2, w_3 = 3, w_4 = 4$ and $w_5 = 5$ so that $\bar{w} = 3$. Let $c = 1 \rightarrow w_0 = \bar{w} = 3$.

First iteration: $w_{1,t} = 2, w_{2,t} = 4, w_{3,t} = 3, w_{4,t} = 3$ and $w_{5,t} = 3$.

Second iteration: $w_{1,t} = 3, w_{2,t} = 3, w_{3,t} = 3, w_{4,t} = 3$ and $w_{5,t} = 3$.

- Typically, the value of c lies between 3 and 6.
- We can also replace $\overline{w}$ by the median of the weights w_k , in which case $w_0 = c\text{Med}(w_k)$.
- Advantage: this method is independent of the survey variables.
- Drawback: the resulting estimator may be efficient for some y -variables but inefficient for others.

Weight trimming

- Another weight trimming method was proposed Potter (1990) who suggested modelling the distribution of the weights.
- He assumed that the sampling weights follow a beta distribution:

$$f(w_k) = \frac{n\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} (1/(nw_k))^{\alpha-1} (1 - 1/(nw_k))^{\beta-1}.$$

- The values of the parameters α and β are estimated using an estimation method (e.g., method of moments or maximum likelihood estimation)
- Rationale: A weight value with a very low probability (say 0.01) of occurring is trimmed.
- The threshold is set to the value w_0 such that $P(w_k \leq w_0) = 0.99$.

- Once the value w_0 is set, the weights are trimmed according to (18).
- This process is repeated a number of times (Potter suggests 10 times), using the newly trimmed weights from the previous iteration.
- Advantage: this method is independent of the survey variables.
- Drawback: the resulting estimator may be efficient for some y -variables but inefficient for others.

Weight trimming

- Typically, weight trimming is performed on the final weights (i.e., after calibration).
- After weight trimming the calibration constraints are not satisfied, in general.
- Solution: the trimmed weights $w_{k,t}$ are modified through some form of calibration. However, some calibrated weights may be larger than the threshold w_0 .
- Better solution: incorporate the weight trimming as part of the calibration process. More specifically, we minimize

$$\sum_{k \in S} \frac{d_k G(\tilde{w}_k / d_k)}{q_k}$$

subject to

$$\sum_{k \in S} \tilde{w}_k \mathbf{x}_k = \mathbf{t}_x \quad \text{and} \quad \tilde{w}_k < w_0.$$

Weight smoothing

- Weight smoothing: introduced by Beaumont (2008).
- Weight smoothing starts by postulating a model linking the design weights to the survey variables:

$$d_k = f(\mathbf{y}_k; \boldsymbol{\gamma}) + \epsilon_k, \quad (19)$$

where $\mathbf{y}_k = (y_{1k}, \dots, y_{pk})^\top$ and $\boldsymbol{\gamma}$ is a p -vector of unknown coefficients.

- Model (19) suggests that a portion of the weight d_k is explained by (some of) the survey variables, while the other portion corresponds to random noise.
- The smoothed weights are given by

$$\hat{d}_k = f(\mathbf{y}_k; \hat{\boldsymbol{\gamma}}),$$

where $\hat{\boldsymbol{\gamma}}$ is a suitable estimator (e.g., the generalized least square estimator) of $\boldsymbol{\gamma}$.

Weight smoothing

- Rationale behind weight smoothing: "get rid" of the portion of the weight d_k corresponding to random noise while preserving the portion which is related to the survey variables.
- The resulting smoothed expansion estimator is

$$\hat{t}_{y,smooth} = \sum_{k \in S} \hat{d}_k y_k.$$

- The properties (e.g., bias) of the smoothed estimator depend on the validity of the postulated model. Classical model selection and validation techniques can be used to determine an appropriate model.
- Unlike weight trimming, weight smoothing does not require the determination of an optimal threshold.

Chapter 7: Why a single weighting system?

The culture of a single weighting system

- Recently, the "culture" of a unique weighting system has been criticized; e.g., Salgado et al. (2012) wrote:

Auxiliary information in statistical offices is nowadays abundant, available, up-to-date and of good quality for statistical purposes. We contend that this increasing availability of auxiliary information invites us to consider putting aside the multipurpose property. We reason as follows. From a purely theoretical standpoint, there is no reasoning that supports the multipurpose property. Moreover, adhering to methodological rigour in sampling weights construction, one can easily find reasons not to have a single set of weights.

- In large-scale surveys, there is often a rich source of auxiliary variables, which could be used to produce tailor-made estimates for every parameter of interest.
- Such estimation procedures have the potential to be more efficient than a procedure based on a unique weighting system.

The culture of a single weighting system

- For specific purposes, it is true that tailor-made estimation procedures can be useful.
- This is customary in small area estimation (Rao and Molina, 2015), where it is usually not always possible to obtain estimates using a single set of weights.
- There are other instances for which using specialized estimation procedures may be desirable in terms of efficiency; e.g., Kalberg (2000) and Firth and Bennett (2000).
- For most surveys, the amount of work required for developing specialized estimation procedures for the production of every estimate of interest seems to be a considerable task.
- A single set of weights seems to be necessary to obtain the vast majority of the estimates in a reasonable amount of time.
- In addition, a unique weighting system permits two intuitively appealing properties to be satisfied: (i) the internal consistency property and (i) the external consistency property.

Internal consistency

- Let $\{\tilde{w}_k; k \in S\}$ be the unique weighting system consisting of final weights.
- Assume we have G characteristics of interest, $y_1, \dots, y_G$, and that $y_G = \sum_{g=1}^{G-1} y_g$.
- For instance, y_1 and y_2 may represent the profit and costs of a business, respectively. Its turnover, y_3 , can then be expressed as

$$y_3 = y_1 + y_2.$$

- If we are interested in estimating the population totals $t_{y_g} = \sum_{k \in U} y_{gk}$, $g = 1, \dots, G$, we have the relationship:

$$t_{y_G} = \sum_{k \in U} y_{Gk} = \sum_{k \in U} \sum_{g=1}^{G-1} y_{gk} = \sum_{g=1}^{G-1} t_{y_g}.$$

- This property is often referred to as internal consistency.

Internal consistency

- It seems desirable that this property also holds at the estimate level.
- With a single weighting system $\{\tilde{w}_k; k \in S\}$ applied to $y_1, \dots, y_G$, we obtain G estimates $\hat{t}_{y_g} = \sum_{k \in S} \tilde{w}_k y_{gk}$ satisfying the relationship

$$\hat{t}_{y_G} = \sum_{g=1}^{G-1} \hat{t}_{y_g}.$$

- In this case, we say that the system of estimates $\{\hat{t}_{y_1}, \dots, \hat{t}_{y_g}, \dots, \hat{t}_{y_G}\}$ is internally consistent.
- Had different weighting systems been used for different variables, the internal consistency would have been violated.

External consistency

- In virtually all surveys, estimates are needed for domains.
- A domain is any subpopulation for which we require an estimate.
- For instance, we may be interested in average income by gender or by age group.
- We assume that the population U of size N is divided into D mutually disjoint domains of interest, $U_1, \dots, U_d, \dots, U_D$ of size $N_1, \dots, N_d, \dots, N_D$, respectively. We have

$$\bigcup_{d=1}^D U_d = U \text{ and } \sum_{d=1}^D N_d = N.$$

- Let $S_d = S \cap U_d$ denote the set of sample units falling in domain d .
- Let $t_d = \sum_{k \in U_d} y_k$ be the total of the y -values in the domain d .

External consistency

- Estimating a domain total is easy as t_d can be expressed as

$$t_d = \sum_{k \in U} \delta_k y_k = \sum_{k \in U} \check{y}_k,$$

where δ_k is a domain indicator attached to unit k such that $\delta_k = 1$ if unit k belongs to the domain d and $\delta_k = 0$, otherwise, and $\check{y}_k = \delta_k y_k$.

- If t_y denotes the overall population total, we have the relationship:

$$t_y = \sum_{k \in U} y_k = \sum_{d=1}^D t_d.$$

- That is, the sum over the domains equals the overall population total.
- This property is often referred to as external consistency.

External consistency

- Once again, it seems desirable that this property also holds at the estimate level.
- With a single weighting system $\{\tilde{w}_k; k \in S\}$ applied to the $\check{y}$ -values, we obtain D estimates, which are naturally consistent.
- In the context of domain estimation, it is possible to develop domain-specific weighting systems. Hidiroglou and Patak (2004) showed empirically that domain-specific estimators are often more efficient than those based on a single weighting system. But with domain-specific estimators, the external consistency property is no longer satisfied, in general.
- Conclusion: a single weighting system is generally effective in practice, achieving internal and external consistency and a manageable workload for the survey statisticians.

Chapter 8: Influence and conditional bias

Chapter 8: Influence and conditional bias

- When can weighted estimators go wrong?
- Working models that justify the expansion and calibration estimators
- Definition of influential unit
- Concept of configuration
- Why might influential units be present?
- Definition of the conditional bias
- Special cases
- Estimation of the conditional bias
- Conditional bias for calibration estimators
- Definition of robust estimator

When can weighted estimators go wrong?

- Go back to the case of full response in this second part of the course but influence and robust estimation concepts can also be extended to the case of nonresponse.
- The main goal of weighting methods (expansion or calibration weighting) is to produce estimators with no (asymptotic) design bias.
 - ▶ **Expansion estimator**: design-unbiased
 - ▶ **Calibration estimator**: asymptotically design-unbiased
- Weighted estimators are perfect (no design bias, no design variance) when a certain relationship holds:
 - ▶ **Expansion estimator**: $y_k = \pi_k B$
 - ▶ **Calibration estimator**: $y_k = \mathbf{x}_k^\top \mathbf{B}$

When can weighted estimators go wrong?

- Weighted estimators are thus expected to be efficient when the underlying relationship holds approximately.
- Indeed, weighted estimators can be justified by a **working model** and are "optimal" if the working model holds.
- The presence of **influential values** is an indication that the working model may not be entirely appropriate.
- Influential values do not introduce design bias but they may lead to highly inefficient weighted estimators.
- Robust estimators aim at increasing efficiency often at the expense of introducing a small (hopefully) design bias.

Working model for the expansion estimator

- The expansion estimator $\hat{t}_{y,\pi} = \sum_{k \in S} d_k y_k$ is design-unbiased for $t_y = \sum_{k \in U} y_k$. **This property does not require any model.**
- However, the expansion estimator is **optimal** when the values $y_k, k \in U$, are independently generated by the working model:

$$Y_k = \beta\pi_k + \pi_k\epsilon_k$$

Assumptions:

- (1) $E_m(\epsilon_k | \pi_k) = 0$
- (2) $\text{var}_m(\epsilon_k | \pi_k) = \sigma^2$
- (3) the distribution of ϵ_k is normal

NOTE: “Optimal” means that $\hat{t}_{y,\pi}$ is a design-unbiased estimator (fixed-size design) for which the anticipated variance $E_m \text{var}_p(\hat{t}_{y,\pi})$ reaches the Godambe-Joshi lower bound if assumption (1) holds. If (2) holds, the lower bound reaches its lowest value.

Working model for the calibration estimator

- Calibration estimators $\hat{t}_{y,C} = \sum_{k \in S} \tilde{w}_k y_k$ are design-consistent for t_y . **This property does not require any model.**
- However, calibration estimators are **optimal** when the values $y_k, k \in U$, are independently generated by the working model:

$$Y_k = \mathbf{x}_k^\top \boldsymbol{\beta} + \pi_k \epsilon_k$$

Assumptions:

- (1) $E_m(\epsilon_k | \mathbf{x}_k) = 0$
- (2) $\text{var}_m(\epsilon_k | \mathbf{x}_k) = \sigma^2$
- (3) the distribution of ϵ_k is normal

NOTE: “optimal” here means that $\hat{t}_{y,C}$ is an estimator whose anticipated variance, $E_m \text{var}_p(\hat{t}_{y,C})$, asymptotically reaches the Godambe-Joshi lower bound if assumption (1) is true. If (2) holds, the lower bound reaches its lowest value.

Definition of influential unit

Definition

Influential unit: A unit may be considered influential if omitting it from the population (and from the sample, if selected) has a significant effect on the sampling error.

Example

Consider $\hat{t}_{y,\pi}$, an estimator of t_y , a fixed-sized without replacement π ps design ($\pi_k = \frac{nz_k}{\sum_{l \in U} z_l}$), and define the influence as:

$$\delta_k = (\hat{t}_{y,\pi} - t_y) - (\hat{t}_{y,\pi}^{(-k)} - t_y^{(-k)})$$

$$\hat{t}_{y,\pi}^{(-k)} = \begin{cases} \sum_{l \in s, l \neq k} \frac{y_l}{\pi_l^{(-k)}} \text{ with } \pi_l^{(-k)} = \frac{(n-1)z_l}{(\sum_{j \in U} z_j) - z_k} & , k \in s; \\ \sum_{l \in s} \frac{y_l}{\pi_l^{(-k)}} \text{ with } \pi_l^{(-k)} = \frac{nz_l}{(\sum_{j \in U} z_j) - z_k} & , k \in U - s. \end{cases}$$

Example

Influence of unit k varies with whether it is selected ($k \in s$) or not ($k \in U - s$) and it can be shown that:

$$\delta_k = \begin{cases} \frac{n}{n-1}(d_k - 1)(y_k - \pi_k \hat{\beta}^{ML}) & , k \in s; \\ -(y_k - \pi_k \hat{\beta}^{ML}) & , k \in U - s. \end{cases}$$

- $\hat{\beta}^{ML} = \frac{\hat{t}_{y,\pi}}{n}$ is the Maximum Likelihood estimator of β (see the working model for the expansion estimator)
- The influence of a sampled unit $k \in s$ depends on the weight, d_k , and the residual, $y_k - \pi_k \hat{\beta}^{ML}$.
- If $\pi_k = 1$, then $\delta_k = 0$.
- A non-sampled unit also has an influence on the sampling error.

Example

We can also show that

$$\hat{t}_{y,\pi} - t_y = \sum_{k \in U} \delta_k + \frac{1}{n-1} \sum_{k \in s} (y_k - \pi_k \hat{\beta}^{ML}) \simeq \sum_{k \in U} \delta_k$$

- The influence δ_k can thus be viewed as the contribution unit k makes to the sampling error $\hat{t}_{y,\pi} - t_y$.
- If the working model holds, the most influential units do not raise any issue and, in principle, should not be treated.
- On the other hand, influential units that are the result of a model misspecification may lead to a very inefficient expansion estimator and require some treatment.

The concept of configuration

The influence of a unit k depends on several characteristics, which we call the configuration.

Definition

A configuration consists of the following four elements:

- (1) the variable y and its distribution in the population (e.g., income, age)
- (2) the population parameter θ (e.g., total, mean, median)
- (3) the sampling design and the associated estimator, $\hat{\theta}$, for example
 - ▶ simple random sampling without replacement and the expansion estimator
 - ▶ simple random sampling without replacement and a calibration estimator
 - ▶ stratified simple random sampling without replacement and the expansion estimator
- (4) whether or not unit k is selected in the sample

The concept of configuration

- A unit may be influential in one configuration and have no influence in another
- Simply modifying a single element of the configuration can dramatically change the influence of a unit
- For example, assigning $\pi_k = 1$ to unit k is enough to reduce its influence to 0.
- Expansion and calibration estimators are (approximately) design-unbiased. The potential presence of influential units in the sample does not introduce bias **but it may result in a substantial increase in the variance.**

Why might influential units be present?

Influential units are more likely to appear when the working model m (for which $\hat{\theta}$ is optimal) is not specified correctly.

Some examples that may lead to the potential presence of influential units:

- A **skewed** distribution of the model errors. (It is common for the y variables to be skewed in business surveys, often resulting in skewed errors.)
- Extreme sample weights, d_k , especially if they are associated with large model errors.
- A sampling frame that contains some significant errors.

Why might influential units be present?

Example

Consider the expansion estimator and the model under which it is optimal:

$$\frac{Y_k}{\pi_k} \sim N(\beta, \sigma^2)$$

Also, let $\pi_k \propto z_k$ for a given variable z_k in the sampling frame, such as the number of employees in a firm.

Now assume that z_k is incorrect for one population unit. The value of z_k in the frame is small $\Rightarrow \pi_k$ is small. During collection, we observe that z_k should have been large and thus that π_k should have been large. **The model does not hold for this influential unit: a potentially large error due to a large weight d_k .**

A similar problem arise in stratified simple random without replacement designs. The missclassified units are called “stratum jumpers”.

Definition of the conditional bias

- We have already defined δ_k , a measure of influence for unit k .
- We now introduce another important concept: the conditional bias for unit k .
- They are both closely related.

Definition

Conditional bias: For any estimator $\hat{\theta}$ of θ , the conditional bias for a unit $k \in s$ is defined as:

$$B_k(I_k = 1) = E_p(\hat{\theta} - \theta | I_k = 1)$$

- Interpretation of the conditional bias $B_k(I_k = 1)$: average sampling error over **all samples containing unit k** .
- Conditional bias $B_k(I_k = 1)$: measure of the influence of sample unit k .

Definition of the conditional bias

- It is also possible to define the conditional bias of a non-sampled unit:

$$B_k(I_k = 0) = E_p(\hat{\theta} - \theta | I_k = 0).$$

- Interpretation of the conditional bias $B_k(I_k = 0)$: the average sampling error for all samples that do not contain unit k .
- Conditional bias $B_k(I_k = 0)$: a measure of the influence of a non-sampled unit.
- If $E_p(\hat{\theta}) = \theta$, we can express $B_k(I_k = 0)$ as

$$B_k(I_k = 0) = -\frac{1}{(d_k - 1)} B_k(I_k = 1).$$

Conditional bias: expansion estimator

- Let $t_y = \sum_{k \in U} y_k$ be the population total of y .
- Let $\hat{t}_{y,\pi} = \sum_{k \in S} d_k y_k$ be the expansion (or Horvitz-Thompson) estimator.
- The conditional bias for a sample unit k is:

$$\begin{aligned} B_k^{HT} (I_k = 1) &= E_p(\hat{t}_{y,\pi} - t_y | I_k = 1) \\ &= \sum_{l \in U} \left(\frac{\pi_{kl} - \pi_k \pi_l}{\pi_k \pi_l} \right) y_l. \end{aligned}$$

- In general, the conditional bias depends on the second-order inclusion probabilities, π_{kl} .
- The conditional bias is generally unknown → it must be estimated
- If $\pi_k = 1$ then $B_k^{HT} (I_k = 1) = 0$.

Conditional bias: expansion estimator

- Poisson sampling:

$$B_k^{HT}(I_k = 1) = (d_k - 1)y_k$$

- Sample unit k is very influential if its sampling weight d_k and/or its value y_k is large.
- **NOTE:** $B_k^{HT}(I_k = 0) = -y_k$
- Stratified simple random sampling:

$$k \in s_h : B_k^{HT}(I_k = 1) = \frac{N_h}{(N_h - 1)} \left(\frac{N_h}{n_h} - 1 \right) (y_k - \bar{Y}_h)$$

- Sample unit $k \in s_h$ is very influential if its sampling weight, N_h/n_h , is large and/or its value y_k is far from the stratum mean, $\bar{Y}_h$.
- **NOTE:** $k \in U_h - s_h : B_k^{HT}(I_k = 0) = -\frac{N_h}{N_h - 1} (y_k - \bar{Y}_h)$

Conditional bias: expansion estimator

- High-entropy π ps without replacement sampling:

$$B_k^{HT}(I_k = 1) \approx (d_k - 1)(y_k - B\pi_k),$$

where $B = \{\sum_{l \in U} \pi_l(1 - \pi_l)\}^{-1} \sum_{l \in U} (1 - \pi_l)y_l$.

- Expression obtained using the general formula and the Hajek approximation of π_{kl}
- **NOTE:** B can be interpreted as an estimate of the model parameter β from the following model obtained from the following model:

$$Y_k = \beta\pi_k + \epsilon_k,$$

where $E_m(\epsilon_k|\pi_k) = 0$ and $\text{var}_m(\epsilon_k|\pi_k) = \sigma^2(d_k - 1)^{-1}$.

- Slightly different from the working model justifying the optimality of the expansion estimator.

Conditional bias: expansion estimator

Example

Consider a population of size $N = 100$, such that

$$y_1 = 0, y_2 = \dots = y_{99} = 500 \text{ and } y_{100} = 1,000 \Rightarrow \bar{Y} = 500.$$

Simple random sampling without replacement: When sampled, units 1 and 100 have a large influence because their respective values are far from $\bar{Y} = 500$.

$$B_k^{HT}(I_k = 1) = \frac{N}{(N-1)} \left(\frac{N}{n} - 1 \right) (y_k - \bar{Y})$$

Poisson sampling: Unit 1, if sampled, has no influence while unit 100 has the largest influence.

$$B_k^{HT}(I_k = 1) = (d_k - 1) (y_k - 0)$$

Poisson sampling vs simple random without replacement sampling



- The following decomposition is always true:

$$(\hat{t}_{y,\pi} - t_y) + \left(\sum_{k \in s} d_k A_k - \sum_{k \in U} A_k \right) = \sum_{k \in U} B_k^{HT}, \quad (20)$$

where

$$B_k^{HT} = \begin{cases} B_k^{HT} (I_k = 1) & \text{if } k \in s \\ B_k^{HT} (I_k = 0) & \text{if } k \in U - s \end{cases}$$

and

$$A_k = (1 - \pi_k)^{-1} \sum_{\substack{l \in U \\ l \neq k}} \left(\frac{\pi_{kl} - \pi_k \pi_l}{\pi_l} \right) y_l.$$

Link between conditional bias and sampling error

- If the term $(\sum_{k \in S} d_k A_k - \sum_{k \in U} A_k)$ is small compared to the term $(\hat{t}_{y,\pi} - t_y)$, we can write:

$$\hat{t}_{y,\pi} - t_y \approx \sum_{k \in U} B_k^{HT}$$

- **Interpretation:** The conditional bias B_k^{HT} can be viewed as the contribution unit k has on the sampling error $\hat{t}_{y,\pi} - t_y$.
- This approximation seems to hold for most sampling designs ...

Link between conditional bias and sampling error

- For Poisson sampling, we have:

$$\hat{t}_{y,\pi} - t_y = \sum_{k \in U} B_k^{HT}$$

- For simple random without replacement sampling and high-entropy π ps without replacement sampling, we have:

$$\hat{t}_{y,\pi} - t_y \approx \sum_{k \in U} B_k^{HT}$$

provided that N is sufficiently large.

- Thus, for these designs, B_k^{HT} can be interpreted as the contribution unit k has on the sampling error $\hat{t}_{y,\pi} - t_y$.
- This approximation is the basis for the construction of **robust estimators** that are less impacted by influential units.

Link between conditional bias and variance of $\hat{t}_{y,\pi}$

- For an arbitrary sampling design, the design-variance of $\hat{t}_{y,\pi}$ is given by:

$$\begin{aligned} \text{var}_p(\hat{t}_{y,\pi}) &= \sum_{k \in U} \sum_{l \in U} \left(\frac{\pi_{kl} - \pi_k \pi_l}{\pi_k \pi_l} \right) y_k y_l \\ &= \sum_{k \in U} B_k^{HT} (I_k = 1) y_k \end{aligned}$$

- Each population unit contributes to the variance of the estimator $\hat{t}_{y,\pi}$. **A unit k with a large influence $B_k^{HT} (I_k = 1)$ has a large contribution to the variance of $\hat{t}_{y,\pi}$.**

Estimation of the conditional bias

- In general, the conditional bias, $B_k^{HT}(I_k = 1)$, is unknown because it is written as a sum over U :

$$B_k^{HT}(I_k = 1) = \sum_{l \in U} \left(\frac{\pi_{kl} - \pi_k \pi_l}{\pi_k \pi_l} \right) y_l$$

- **Exception:** Poisson sampling, where $B_k^{HT}(I_k = 1)$ reduces to

$$B_k^{HT}(I_k = 1) = (d_k - 1)y_k$$

- **How to estimate $B_k^{HT}(I_k = 1)$?**
- A natural approach is to look for an estimator $\hat{B}_k^{HT}(I_k = 1)$ that is conditionally unbiased for $B_k^{HT}(I_k = 1)$ in the sense that:

$$E_p \left\{ \hat{B}_k^{HT}(I_k = 1) | I_k = 1 \right\} = B_k^{HT}(I_k = 1)$$

Estimation of the conditional bias

- Conditional selection probability: $Pr(I_l = 1 \mid I_k = 1) = \frac{\pi_{kl}}{\pi_k}$
- A conditionally-unbiased estimator of $B_k^{HT} (I_k = 1)$ is given by:

$$\hat{B}_k^{HT}(I_k = 1) = \sum_{j \in S} \left(\frac{\pi_{kl} - \pi_k \pi_l}{\pi_l \pi_{kl}} \right) y_l \quad (21)$$

- Obtained by weighting by the inverse conditional selection probability
- Applicable only if $\pi_{kl} > 0$
- In the special case of stratified simple random sampling, for $k \in s_h$, equation (21) becomes:

$$\hat{B}_k^{HT}(I_k = 1) = \frac{n_h}{n_h - 1} \left(\frac{N_h}{n_h} - 1 \right) (y_k - \bar{y}_h) = \delta_k,$$

where $\bar{y}_h$ is the sample mean in stratum h .

Estimation of the conditional bias

- For some designs, we could also consider a robust estimate of the conditional bias. In the case of a stratified simple random sampling, we could choose:

$$\widehat{B}_k^{HT}(I_k = 1) = \frac{n_h}{n_h - 1} \left(\frac{N_h}{n_h} - 1 \right) (y_k - m_h),$$

where m_h is the sample median in stratum h .

- In reality, whether we estimate the conditional bias in a robust fashion or not, it is of little importance. Both approaches yield robust estimators with very similar properties in terms of bias and variance. This will be illustrated empirically in the next chapter.
- The conditionally-unbiased estimator (21) is simple to compute, can be applied with any sampling design (if π_k and $\pi_{k|}$ are known) and is related to the influence measure δ_k .

Conditional bias: calibration estimators

- For a sample unit k , the conditional bias of a calibration estimator $\hat{t}_{y,C} = \sum_{k \in S} \tilde{w}_k y_k$ of t_y is defined as:

$$B_k^{CAL}(I_k = 1) = E_p(\hat{t}_{y,C} - t_y | I_k = 1)$$

- An exact expression is difficult to obtain because $\hat{t}_{y,C}$ is not **linear** (not a linear function of I_k 's).
- However, for any calibration estimator, if the sample size n is sufficiently large, a first-order Taylor expansion yields:

$$\hat{t}_{y,C} - t_y \approx \sum_{k \in S} d_k E_k - \sum_{k \in U} E_k, \quad (22)$$

where $E_k = y_k - \mathbf{x}_k^T \mathbf{B}$ represents the linearized variable for unit k .

- The right-hand side of this approximation is linear and can be used to approximate the conditional bias $B_k^{CAL}(I_k = 1)$.

- The conditional bias for sample unit k is approximated by:

$$\begin{aligned} B_k^{CAL}(I_k = 1) &= E_p(\hat{t}_{y,C} - t_y | I_k = 1) \\ &\approx E_p\left(\sum_{k \in S} d_k E_k - \sum_{k \in U} E_k \mid I_k = 1\right) \\ &= \sum_{I \in U} \left(\frac{\pi_{kI} - \pi_k \pi_I}{\pi_k \pi_I}\right) E_I. \end{aligned}$$

- Recall:** For the expansion estimator, the conditional bias $B_k^{HT}(I_k = 1)$ is given by the above expression with y_k rather than E_k .

Conditional bias: calibration estimators

- Poisson sampling:

$$B_k^{CAL}(I_k = 1) \approx (d_k - 1)E_k$$

- Simple random sampling without replacement:

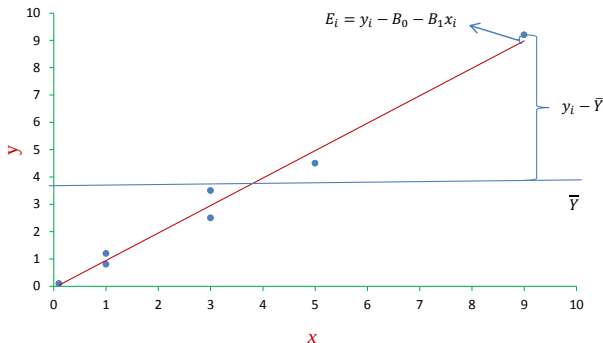
$$B_k^{CAL}(I_k = 1) \approx \frac{N}{(N-1)} \left(\frac{N}{n} - 1 \right) (E_k - \bar{E})$$

- In some situations, we have $\bar{E} = 0$ (e.g., when an intercept is included in the vector $\mathbf{x}_k$ and $q_k = 1$). In this case, the influence is similar to Bernoulli sampling.
- **Conclusion:** A unit may have a large influence if the estimator $\hat{t}_{y,\pi}$ is used and very little influence if the estimator $\hat{t}_{y,C}$ is used, and vice versa.

Influence: expansion estimator vs. calibration estimator

- Consider simple random sampling without replacement
- Vector of auxiliary variables:

$$\mathbf{x}_k = (1, x_k)^\top \Rightarrow \mathbf{t}_\mathbf{x} = \sum_{k \in U} \mathbf{x}_k = (N, t_x)^\top$$



Estimation of the conditional bias

- A conditionally-unbiased estimator of $B_k^{CAL} (I_k = 1)$ is given by:

$$\hat{B}_k^{CAL}(I_k = 1) = \sum_{j \in S} \left(\frac{\pi_{kl} - \pi_k \pi_l}{\pi_l \pi_{kl}} \right) e_l \quad (23)$$

where $e_l = y_l - \mathbf{x}_l^\top \hat{\mathbf{B}}$ and $\hat{\mathbf{B}}$ is an estimator of $\mathbf{B}$.

- Same thing as $\hat{B}_k^{HT} (I_k = 1)$ except that e_l replaces y_l .
- In the special case of stratified simple random sampling, for $k \in s_h$:

$$\hat{B}_k^{CAL}(I_k = 1) = \frac{n_h}{n_h - 1} \left(\frac{N_h}{n_h} - 1 \right) (e_k - \bar{e}_h)$$

Robust estimation

Model m : Working model under which a design-consistent estimator $\hat{\theta}$ of θ is optimal (e.g., expansion or calibration estimator of t_y)

Definition

Robust estimator: An estimator $\hat{\theta}^R$ is robust for θ if:

- (1) it might be significantly more efficient than the weighted estimator $\hat{\theta}$ in the presence of a **small deviation** from the model m ;
- (2) its efficiency is similar to that of $\hat{\theta}$ when model m holds.

Note: “small deviation” may mean that:

- A small proportion of the data is generated by a model other than m (outliers).
- The first two moments of the model m are valid, but the distribution of the errors is highly skewed rather than normal (extreme values in the errors).

Chapter 9: Robust estimation based on the conditional bias

- Robust estimation
- Implementation of the robust estimator
- Robust version of a calibration estimator
- Choice of c
- Simulation studies
- Estimation of the mean squared error

How construct a robust estimator based on the conditional bias?

First, we note that:

$$\hat{t}_{y,\pi} = \left(\hat{t}_{y,\pi} - \sum_{k \in U} \hat{B}_k^{HT} \right) + \sum_{k \in U} \hat{B}_k^{HT}$$

The term in parentheses is (approximately) equal to θ and is thus not affected by influential values.

Ideally, the impact influential units in the population have on the second term would be reduced, and the following estimator used:

$$\hat{t}_{y,\pi}^R = \hat{t}_{y,\pi} - \sum_{k \in U} \hat{B}_k^{HT} + \sum_{k \in U} \psi \left(\hat{B}_k^{HT}; c \right)$$

NOTE: The impact of non-sampled units cannot be reduced because it is impossible to estimate their conditional bias.

Robust estimation

- The underlying idea is thus to reduce the impact of the most influential units **in the sample** on the second term. This yields the robust estimator:

$$\hat{t}_{y,\pi}^R = \hat{t}_{y,\pi} - \sum_{k \in U} \hat{B}_k^{HT} + \sum_{k \in S} \psi \left(\hat{B}_k^{HT}(I_k = 1); c \right) + \sum_{k \in U-S} \hat{B}_k^{HT}(I_k = 0).$$

- This estimator simplifies to:

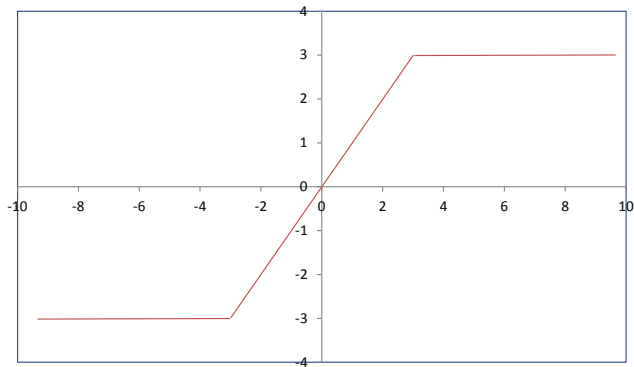
$$\hat{t}_{y,\pi}^R = \hat{t}_{y,\pi} - \sum_{k \in S} \hat{B}_k^{HT}(I_k = 1) + \sum_{k \in S} \psi \left(\hat{B}_k^{HT}(I_k = 1); c \right)$$

- Role of the function ψ : reduce the influence of the most influential units.
- Example of the ψ function: the Huber function.

$$\psi_c(t) = \begin{cases} c & \text{if } t > c \\ t & \text{if } |t| \leq c \\ -c & \text{if } t < -c \end{cases}$$

Huber function $\psi_3(t)$ (with $c = 3$)

- Remark: $0 \leq \frac{\psi_c(t)}{t} \leq 1$.



$$\hat{t}_{y,\pi}^R = \hat{t}_{y,\pi} - \sum_{k \in S} \hat{B}_k^{HT} (I_k = 1) + \sum_{k \in S} \psi \left(\hat{B}_k^{HT} (I_k = 1); c \right)$$

- $\hat{t}_{y,\pi}^R$ is Cochran-consistent. In other words, when $S = U$, we have $\hat{t}_{y,\pi}^R = t_y$.
- When $c \rightarrow \infty$, the robust estimator $\hat{t}_{y,\pi}^R$ converges to the non-robust estimator $\hat{t}_{y,\pi}$.
- In the stratified simple random without replacement case, $\hat{t}_{y,\pi}^R$ coincides with the Kokic and Bell (1994) estimator if we ignore the factor $N_h/(N_h - 1)$.

Implementation of the robust estimator: re-weighted form

- For an arbitrary sampling design, the robust estimator $\hat{t}_{y,\pi}^R$ can be written as

$$\hat{t}_{y,\pi}^R = \sum_{k \in S} \tilde{d}_k y_k,$$

where

$$\tilde{d}_k = d_k - \alpha_k \frac{\hat{B}_k^{HT}(I_k = 1)}{y_k}$$

and

$$\alpha_k = 1 - \frac{\psi\left(\hat{B}_k^{HT}(I_k = 1)\right)}{\hat{B}_k^{HT}(I_k = 1)}.$$

- Recall: $0 \leq \psi(t)/t \leq 1 \Rightarrow 0 \leq \alpha_k \leq 1$.
- The weights $\tilde{d}_k$ are a function of the y -values $\Rightarrow$ one weight per y -variable!

Implementation of the robust estimator: re-weighted form

- Poisson sampling

$$\tilde{d}_k = \alpha_k + (1 - \alpha_k)d_k \Rightarrow 1 \leq \tilde{d}_k \leq d_k$$

- Simple random sampling without replacement (with $N/(N-1) = 1$):

$$\tilde{d}_k = \gamma_k + \{1 - \gamma_k\} \frac{N}{n},$$

where $\gamma_k = \alpha_k \left(\frac{y_k - \bar{y}_U}{y_k} \right)$.

- ▶ If $\alpha_k = 0 \Rightarrow \gamma_k = 0 \Rightarrow \tilde{d}_k = N/n$;
- ▶ If $\alpha_k = 1$

$$\Rightarrow \tilde{d}_k = 1 + \frac{\left(\frac{N}{n} - 1 \right)}{\left(\frac{y_k}{\bar{y}_U} \right)}.$$

Implementation of the robust estimator: imputed form

- For an arbitrary sampling design, the robust estimator $\hat{t}_{y,\pi}^R$ can be written as

$$\hat{t}_{y,\pi}^R = \sum_{k \in S} d_k \tilde{y}_k,$$

where

$$\tilde{y}_k = y_k - \alpha_k \frac{\hat{B}_k^{HT}(I_k = 1)}{d_k}.$$

- Poisson sampling: $\tilde{y}_k = y_k (1 - \alpha_k (1 - \pi_k)) \Rightarrow \frac{y_k}{d_k} \leq \tilde{y}_k \leq y_k$
- Simple random sampling without replacement:

$$\tilde{y}_k = \gamma_k \bar{y}_U + (1 - \gamma_k) y_k,$$

where $\gamma_k = \alpha_k(1 - f)$.

- If $\alpha_k = 1$ then $\tilde{y}_k = f y_k + (1 - f) \bar{y}_U \Rightarrow$ type II winsorization

Robust version of the calibration estimator

- The same line of reasoning that yielded the robust version of the Horvitz-Thompson estimator can be used to obtain a robust version of the calibration estimator:

$$\hat{t}_{y,C}^R = \hat{t}_{y,C} - \sum_{k \in S} \hat{B}_k^C(I_k = 1) + \sum_{k \in S} \psi\left(\hat{B}_k^C(I_k = 1); c\right)$$

- In the case of a Poisson sampling or a simple random sampling without replacement (with $N/(N-1) = 1$ and $\bar{E}_U = 0$), the conditional bias of unit k with respect to $\hat{t}_{y,C}$ is given by

$$B_k^C(I_k = 1) \approx (d_k - 1) E_k,$$

where $E_k = y_k - \mathbf{x}_k^\top \mathbf{B}$ and

$$\mathbf{B} = \left(\sum_{k \in U} \mathbf{x}_k q_k \mathbf{x}_k^\top \right)^{-1} \sum_{i \in U} \mathbf{x}_i q_i y_i.$$

Robust version of the calibration estimator

- Let $\tilde{\mathbf{B}}$ represent the estimator for the coefficient $\mathbf{B}$ in the term for the conditional bias. For example, $\tilde{\mathbf{B}}$ may be a robust estimator (e.g., the M estimator) or the non-robust estimator $\hat{\mathbf{B}}$.
- The robust version of the generalized regression estimator can be written as:

$$\begin{aligned}\hat{t}_{y,c}^R &= \hat{t}_{y,c} - \sum_{k \in S} \hat{B}_k^C(I_k = 1) + \sum_{k \in S} \psi\left(\hat{B}_k^{REG}(I_k = 1); c\right) \\ &= \sum_{k \in S} y_k + \sum_{k \in U-S} \mathbf{x}_k^\top \tilde{\mathbf{B}} + (\mathbf{t}_x - \hat{\mathbf{t}}_{x,\pi})^\top (\hat{\mathbf{B}} - \tilde{\mathbf{B}}) \quad (24) \\ &\quad + \sum_{k \in S} \psi\left(\hat{B}_k^C(I_k = 1); c\right).\end{aligned}$$

Robust version of the calibration estimator

- The term $(\mathbf{t}_x - \hat{\mathbf{t}}_{x,\pi})^\top (\hat{\mathbf{B}} - \tilde{\mathbf{B}})$ in (24) can be viewed as a correction term. If the estimator $\tilde{\mathbf{B}}$ is very robust, then the correction term will likely yield a less biased estimator.
- If $\tilde{\mathbf{B}} = \hat{\mathbf{B}}$, this estimator simplifies to

$$\hat{t}_{y,C}^R = \sum_{k \in S} y_k + \sum_{k \in U-S} \mathbf{x}_k^\top \hat{\mathbf{B}} + \sum_{k \in S} \psi \left(\hat{B}_k^C(I_k = 1); c \right).$$

- In this case, the form of $\hat{t}_{y,C}^R$ is similar to that in Chambers (1986); see also Beaumont and Rivest (2009).

Choice of c

Let $\hat{\theta}^R(c) = \hat{\theta} - \sum_{k \in S} \hat{B}_k(I_k = 1) + \sum_{k \in S} \psi(\hat{B}_k(I_k = 1); c)$ be a robust estimator of θ

If c is sufficiently large and we use the Huber function, then $\hat{\theta}^R(c) = \hat{\theta}$, which is (approximately) unbiased for a certain normal model

If $c = 0$ then the estimator $\hat{\theta}^R(0)$ may be very biased, but possibly more stable

Choice of c : We find the value of c that minimizes an estimate:

- of the MSE of $\hat{\theta}^R(c)$
- of the largest conditional bias in the sample $\hat{\theta}^R(c)$

Minimization of an estimate of the MSE of $\hat{\theta}^R(c)$:

- Estimation of the MSE may be complicated $\rightarrow$ Bootstrap may be useful (we will return to this)
- Can be obtained by considering different values of c and choosing the one that yields the smallest MSE or by using a Newton-Raphson type algorithm (with derivatives computed numerically).
- **Problem:** The global minimum is not always easy to find.
- Processing time must also be considered

Minimization of an estimate of the largest conditional bias in the sample of $\hat{\theta}^R(c)$

- Easier to implement and requires less computer time

Choice of c : minimize the largest conditional bias

- Consider the class of robust estimators:

$$\hat{t}_{y,\pi}^R(c) = \hat{t}_{y,\pi} + \Delta(c),$$

where $\Delta(c)$ is a random variable.

- Several robust estimators we have already covered belong to this class.
- We are seeking the value of $\Delta(c)$ that minimizes

$$\max_{k \in S} \left\{ |\hat{B}_k^{RHT}(I_k = 1)| \right\},$$

where $\hat{B}_k^{RHT}(I_k = 1)$ represents the conditional bias of unit i with respect to $\hat{t}_{y,\pi}^R(c)$

Choice of c : minimize the largest conditional bias

- We have

$$B_k^{RHT}(I_k = 1) = E_p\{\hat{t}_{y,\pi}^R(c)|I_k = 1\} - \theta = B_k^{HT}(I_k = 1) + E_p(\Delta(c)|I_k = 1)$$

- The estimator of $B_k^R(I_k = 1)$:

$$\hat{B}_k^{RHT}(I_k = 1) = \hat{B}_k^{HT}(I_k = 1) + \Delta(c)$$

- $E_p\left(\hat{B}_k^{RHT}(I_k = 1)\right) = B_k^{RHT}(I_k = 1) \rightarrow \hat{B}_k^{RHT}(I_k = 1)$ is conditionally unbiased for $B_k^{RHT}(I_k = 1)$
- Beaumont, Haziza and Ruiz-Gazen (2013): the value $\Delta(c)$ that minimizes $\max_{k \in S} \left\{ |\hat{B}_k^{RHT}(I_k = 1)| \right\}$ is given by

$$\Delta(c_{opt}) = -\frac{1}{2}(\hat{B}_{min}^{HT} + \hat{B}_{max}^{HT}),$$

where $\hat{B}_{min}^{HT} = \min_{k \in S} \left(\hat{B}_k^{HT}(I_k = 1) \right)$ and

$$\hat{B}_{max}^{HT} = \max_{k \in S} \left(\hat{B}_k^{HT}(I_k = 1) \right).$$

Choice of c : minimize the largest conditional bias

- This principle leads to the robust estimator

$$\begin{aligned}\hat{t}_{y,\pi}^R(c_{opt}) &= \hat{\theta}^{HT} + \Delta(c_{opt}) \\ &= \hat{t}_{y,\pi} - \frac{1}{2}(\hat{B}_{min}^{HT} + \hat{B}_{max}^{HT}).\end{aligned}\quad (25)$$

Remarks on the robust estimator:

- Design consistent: $\hat{t}_{y,\pi}^R(c_{opt}) - \theta = O_p(Nn^{-1/2})$.
- Very easy to compute (compared to minimizing the estimated mean square error)

Choice of c : minimize the largest conditional bias

- The value c_{opt} is obtained by solving for c :

$$\begin{aligned}\Delta(c) &= \sum_{k \in S} \hat{B}_k(I_k = 1) - \sum_{k \in S} \psi\left(\hat{B}_k(I_k = 1); c\right) \\ &= \frac{1}{2} \left(\hat{B}_{min} + \hat{B}_{max} \right)\end{aligned}$$

NOTE: If the Huber function is used there is always a solution to this equation, but it may not be unique. In this event, the largest value of c that satisfies the equation is chosen; see Beaumont, Haziza and Ruiz-Gazen (2013) and Favre Martinoz, Haziza and Beaumont (2015).

Choice of c : minimize the largest conditional bias

Unit	c_1	c_2	...	c_q
	$\hat{t}_{y,\pi}^R(c_1)$	$\hat{t}_{y,\pi}^R(c_2)$	...	$\hat{t}_{y,\pi}^R(c_q)$
1	$\hat{B}_1^{RHT}(I_1 = 1)(c_1)$	$\hat{B}_1^{RHT}(I_1 = 1)(c_2)$	...	$\hat{B}_1^{RHT}(I_1 = 1)(c_q)$
2	$\hat{B}_2^{RHT}(I_2 = 1)(c_1)$	$\hat{B}_2^{RHT}(I_2 = 1)(c_2)$	...	$\hat{B}_2^{RHT}(I_2 = 1)(c_q)$
...	...	...	...	...
...	...	...	...	...
n	$\hat{B}_n^{RHT}(I_n = 1)(c_1)$	$\hat{B}_n^{RHT}(I_n = 1)(c_2)$	...	$\hat{B}_n^{RHT}(I_n = 1)(c_q)$
	$\max \left\{ \hat{B}_k^{RHT}(I_k = 1)(c_1) \right\}$	$\max \left\{ \hat{B}_k^{RHT}(I_k = 1)(c_2) \right\}$	...	$\max \left\{ \hat{B}_k^{RHT}(I_k = 1)(c_q) \right\}$

- In the case of simple random sampling without replacement, the (25) estimator can be written as

$$\widehat{t}_{y,\pi}^R(c_{opt}) = \sum_{k \in S} \widetilde{d}_k y_k,$$

where $\widetilde{d}_k = \frac{N-1}{n-1} - \frac{N-n}{2(n-1)}$ for the smallest and largest units in the sample, and $\widetilde{d}_k = \frac{N-1}{n-1}$ for all other units in the sample. Observe that $\sum_{k \in S} \widetilde{d}_k = N$.

- For example, if $N = 1,000$ and $n = 50$, we have $d_k = 20$ for all k , $\widetilde{d}_k = 10.69$ for the smallest and largest units in the sample, and $\widetilde{d}_k = 20.38$ for all other units in the sample.

- In the case of a Bernoulli design (with inclusion probability π), the estimator (25) can be written

$$\widehat{t}_{y,\pi}^R(c_{opt}) = \sum_{k \in S} \widetilde{d}_k y_k,$$

where $\widetilde{d}_k = \frac{1}{2} \frac{1-\pi}{\pi}$ for the smallest and largest units in the sample, and $\widetilde{d}_k = \pi$ for all other units in the sample.

- For example, if $N = 1,000$ and $\pi = 0.05$, we have $d_k = 20$ for all k , $\widetilde{d}_k = 10.5$ for the smallest and largest units in the sample, and $\widetilde{d}_k = 20$ for all other units in the sample.

Simulation study 1

- We generated a population of size $N = 5,000$ with variable of interest y
 - ▶ The population was generated from a mixture of two lognormal distributions:

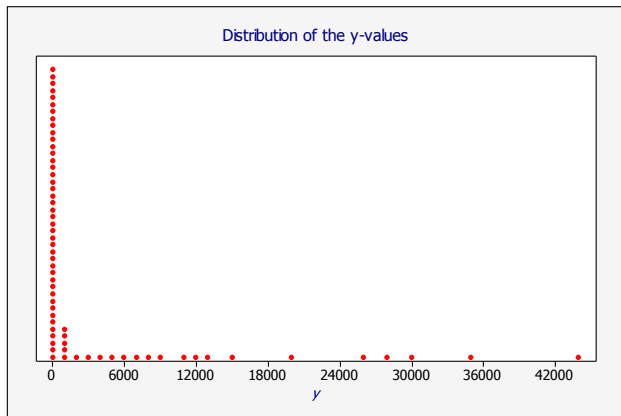
$$y_0 \sim LN(5, 1) \quad \text{and} \quad y_1 \sim LN(8, 2)$$

$$y_k = (1 - \delta_k)y_{0k} + \delta_k y_{1k} \quad \text{and} \quad Prob(\delta_k = 1) = 0.01$$

- In each population, $R = 25,000$ samples of size $n = 50; 100; 500$ were drawn using a simple random sampling without replacement.

Simulation study 1

Dot plot of the y -values



Simulation study 1

We computed 3 estimators:

- the expansion estimator: $\hat{t}_{y,\pi} = \sum_{k \in S} d_k y_k$
- the robust estimator with the constant c chosen according to the min-max criterion:

$$\hat{t}_{y,\pi}^R = \hat{t}_{y,\pi} - \frac{1}{2} \left\{ \min \hat{B}_k^{HT}(I_k = 1) + \max \hat{B}_k^{HT}(I_k = 1) \right\}$$

- ▶ with a non-robust estimate of the conditional bias:

$$\hat{B}_k^{HT}(I_k = 1) = \frac{n}{(n-1)} \left(\frac{N}{n} - 1 \right) (y_k - \bar{y});$$

- ▶ with a robust estimate of the conditional bias:

$$\hat{B}_k^{HT}(I_k = 1) = \frac{n}{(n-1)} \left(\frac{N}{n} - 1 \right) (y_k - m).$$

Monte Carlo Measures

- Monte Carlo relative bias (in percentage):

$$RB_{MC}(\hat{\theta}) = 100 \times [E_{MC}(\hat{\theta}) - \theta] / \theta.$$

- Monte Carlo relative efficiency (in percentage) with respect to the expansion estimator:

$$RE = 100 \times MSE_{MC}(\hat{\theta}) / MSE_{MC}(\hat{\theta}^{HT}).$$

Simulation study 1

n	$\hat{t}_{y,\pi}$	$\hat{t}_{y,\pi}^R$ (non robust)	$\hat{t}_{y,\pi}^R$ (robust)
50	0.11 (100)	-12.22 (44.0)	-13.01 (43.2)
100	0.16 (100)	-10.27 (50.7)	-10.70 (50.5)
500	0.01 (100)	-5.50 (79.2)	-5.60 (79.3)

Simulation study 2

- We generated 6 populations with two variables: x and y
- In each population, the values for x were generated from a Gamma distribution with mean 50 and variance 500.
- In the 3 first populations of size 500, 1,000 and 5,000, respectively, the variable y was generated using a ratio model

$$y_k = 2x_k + 15\varepsilon_k,$$

where the errors ε_k were normally distributed with mean 0 and variance σ^2 . The coefficient of correlation between x and y was set to approximately 0.8.

- These populations contained no outliers.

Simulation study 2

- In the 3 last populations of size 500, 1,000 and 5,000, respectively, the variable y was generated using a mixture

$$y_k = \delta_k(2x_k + 15\varepsilon_k) + (1 - \delta_k)u_k,$$

where the variable u was generated from a normal distribution with mean 1,200 and variance 40,000 and the variables δ_k was generated independently from a Bernoulli distribution with parameter $p = 0.01$.

- Therefore, the proportion of outliers in these populations was approximately equal to 1%.
- From each population we drew $R = 10,000$ samples using a Poisson design with inclusion probabilities π_k , proportional to x_k ; i.e., $\pi_k = \tilde{n}x_k / \sum_{i \in U} x_i$, where $\tilde{n}$ represents the expected sample size.
- The expected sampling fraction, $\tilde{n}/N$, was set to 0.02 and 0.10.

Simulation study 2

We computed 3 estimators:

- the expansion estimator: $\hat{t}_{y,\pi} = \sum_{k \in S} d_k y_k$
- the robust estimator

$$\hat{t}_{y,\pi}^R = \hat{t}_{y,\pi} - \frac{1}{2} \left\{ \min \hat{B}_k^{HT}(I_k = 1) + \max \hat{B}_k^{HT}(I_k = 1) \right\}$$

with the tuning constant c chosen

- ▶ according the min-max criterion;
- ▶ to minimize the estimated MSE of $\hat{t}_{y,\pi}^R$. In the case of Poisson sampling, we have

$$\begin{aligned} mse_P(\hat{t}_{y,\pi}^R) &= \sum_{k \in S} (1 - \pi_k) [y_k + \psi\{B_k^{HT}(I_k = 1)\}]^2 \\ &+ \max \left(0, (\hat{t}_{y,\pi}^R - \hat{t}_{y,\pi})^2 \right. \\ &\left. - \sum_{k \in S} (1 - \pi_k) [\psi\{B_k^{HT}(I_k = 1)\} - B_k^{HT}(I_k = 1)]^2 \right). \end{aligned}$$

Simulation study 2

f	$\tilde{n}$	$\hat{t}_{y,\pi}$	$\hat{t}_{y,\pi}^R(cb)$	$\hat{t}_{y,\pi}^R(mse)$
Populations without outliers				
0.02	10	-0.2 (100)	-10.0 (108)	-11.4 (109)
	20	0.2 (100)	-4.7 (103)	-5.8 (104)
	100	-0.0 (100)	-1.00 (100)	-1.4 (101)
0.1	50	-0.1 (100)	-1.9 (101)	-2.4 (102)
	100	-0.1 (100)	-1.0 (100)	-1.3 (101)
	500	0.0 (100)	-0.1 (100)	-0.2 (100)
Populations with outliers				
0.02	10	-0.1 (100)	-15.2 (55)	-23.2 (43)
	20	-0.2 (100)	-10.0 (68)	-14.8 (62)
	100	-0.0 (100)	-5.3 (93)	-8.4 (103)
0.1	50	0.1 (100)	-7.2 (70)	-11.5 (70)
	100	-0.0 (100)	-4.4 (89)	-6.6 (95)
	500	0.0 (100)	-1.4 (101)	-2.1 (104)

Simulation study 2

We also computed 3 versions of the ratio estimator:

- The usual ratio estimator, $\hat{t}_{y,C}$.
- A robust version of the ratio estimator, where c was determined according to the min-max criterion. This yields, $\hat{t}_{y,C}^R = \hat{t}_{y,C} - \frac{1}{2} \left\{ \min \hat{B}_k^C(I_k = 1) + \max \hat{B}_k^C(I_k = 1) \right\}$.
- The conditional bias appearing in the calculation of $\hat{t}_{y,C}^R$ is estimated with:
 - ▶ a non-robust estimate

$$\hat{B}_k^C(I_k = 1) = \frac{n}{(n-1)} \left(\frac{N}{n} - 1 \right) (y_k - \hat{B}x_k).$$

- ▶ a robust estimate

$$\hat{B}_k^C(I_k = 1) = \frac{n}{(n-1)} \left(\frac{N}{n} - 1 \right) (y_k - \hat{B}_R x_k).$$

- ▶ $\hat{B}_R$: *M*-estimator computed using the Huber function with $c = 1.345$.

Simulation study 2

f	n	$\hat{t}_{y,C}$	$\hat{t}_{y,C}^R(nr)$	$\hat{t}_{y,C}^R(r)$
Populations without outliers				
0.02	10	-0.4 (100)	-0.0 (97)	0.0 (101)
	20	0.2 (100)	-0.0 (95)	-0.0 (98)
	100	0.1 (100)	-0.0 (98)	-0.1 (100)
0.1	50	0.1 (100)	-0.0 (97)	0.0 (100)
	100	-0.0 (100)	-0.0 (97)	-0.0 (99)
	500	-0.0 (100)	-0.0 (100)	-0.2 (100)
Populations with outliers				
0.02	10	-0.4 (100)	-6.2 (52)	-7.0 (45)
	20	-0.0 (100)	-5.1 (52)	-5.8 (46)
	100	0.1 (100)	-4.1 (88)	-4.4 (87)
0.1	50	0.0 (100)	-5.5 (59)	-5.8 (58)
	100	0.1 (100)	-3.4 (79)	-3.6 (78)
	500	0.0 (100)	-1.2 (100)	-1.3 (100)

Estimation of the mean squared error

If $E_p(\hat{\theta}) \simeq \theta$, the MSE of $\hat{\theta}^R$ can be written:

$$\begin{aligned} \text{MSE}_p(\hat{\theta}^R) &= E_p(\hat{\theta}^R - \theta)^2 \\ &= V_p(\hat{\theta}^R) + \left[E_p(\hat{\theta}^R - \theta) \right]^2 \\ &\simeq V_p(\hat{\theta}^R) + \left[E_p(\hat{\theta}^R - \hat{\theta})^2 - V_p(\hat{\theta}^R - \hat{\theta}) \right] \end{aligned}$$

Gwet and Rivest (1992) suggest the estimator:

$$\text{mse}_p(\hat{\theta}^R) = \hat{V}(\hat{\theta}^R) + \max \left[0, (\hat{\theta}^R - \hat{\theta})^2 - \hat{V}(\hat{\theta}^R - \hat{\theta}) \right]$$

$\hat{\theta}^R$ may have quite a complicated expression, especially when conditional biases have to be estimated. **How are the variances estimated?** Replication methods like the generalized bootstrap are the natural choice.

Linear case:

- We wish to estimate the variance of $\hat{t}_{y,\pi} = \sum_{k \in S} d_k y_k$.
- An estimator of $V_p(\hat{t}_{y,\pi})$ has usually the following quadratic form:

$$\hat{V}(\hat{t}_{y,\pi}) = \sum_{k \in S} \sum_{l \in S} \lambda_{kl} (d_k y_k) (d_l y_l).$$

For example, $\lambda_{kl} = (\pi_{kl} - \pi_k \pi_l) / \pi_{kl}$

- Conditional on the sample s , we randomly generate bootstrap weights d_k^* .
- The bootstrap version of $\hat{t}_{y,\pi}$ is

$$\hat{t}_{y,\pi}^* = \sum_{k \in S} d_k^* y_k$$

How are the bootstrap weights d_k^* generated? They must capture the two first moments:

1. $E_*(\hat{t}_{y,\pi}^*) = \hat{t}_{y,\pi} \longrightarrow E_p(\hat{t}_{y,\pi}) = t_y$
2. $V_*(\hat{t}_{y,\pi}^*) = \hat{V}(\hat{t}_{y,\pi}) \longrightarrow V_p(\hat{t}_{y,\pi})$

These conditions are met if:

1. $E_*(d_k^*) = d_k$
2. $cov_*(d_k^*, d_l^*) = \lambda_{kl} d_k d_l$

NOTE: The bootstrap version of $\sum_{k \in S} z_k$ is $\sum_{k \in S} d_k^* \pi_k z_k$

- Most of the bootstrap methods found in the literature can be written in a weighted form and are special cases of the generalized bootstrap.
- For example: the Rao and Wu bootstrap (1988)
- References: Bertail and Combris (1997); Beaumont and Patak (2012)

Nonlinear case:

- If the statistic of interest can be linearized the bootstrap remains valid.
- This works for smooth functions of totals, parameters defined by estimating equations, quantiles, etc.
- It is not valid when the statistics of interest are extreme.

REMINDER:

$$mse_p(\hat{\theta}^R) = \hat{V}(\hat{\theta}^R) + \max \left[0, (\hat{\theta}^R - \hat{\theta})^2 - \hat{V}(\hat{\theta}^R - \hat{\theta}) \right]$$

The greatest challenge is to bootstrap $\hat{\theta}^R$.

Consider 3 cases:

- **c fixed**: no problem
- **c minimizing** $\max \left(|\hat{B}_k(\hat{\theta}^R(c), I_k = 1)| | k \in S \right)$: bootstrap does not work due to $\hat{B}_{min}$ and $\hat{B}_{max}$
- **c minimizing** $mse_p(\hat{\theta}^R(c))$: requires a double bootstrap if bootstrap is used to estimate the MSE: How do we do this?

In the two last cases, we suggest treating c_{opt} as fixed (even though it is not).

THANK YOU!

MERCI!